

MASTER PRODUCT REQUIREMENTS & SOFTWARE DEVELOPMENT LIFECYCLE DOCUMENT

for

AI-Powered Personalized Tourist Guide & Travel Companion

Problem Statement TC-SO1 — AI Personalized Tourist Guide Software

Document Version	1.0
Document Status	Draft — For Review
Prepared By	Joseph — Founder & Product Owner
Date	21 August 2026
Classification	Internal / Confidential

Prepared as the single master reference for product, design, engineering, QA, and delivery.

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1. Document Control

1.1 Document Information

Document Title	Master Product Requirements & Software Development Lifecycle Document
Product Name	AI-Powered Personalized Tourist Guide & Travel Companion
Problem Statement Reference	TC-SO1 — AI Personalized Tourist Guide Software
Version	1.0
Document Status	Draft — For Review
Prepared By	Joseph (Founder / Product Owner)
Reviewed By	TBC — To Be Confirmed
Approved By	TBC — To Be Confirmed
Date	21 August 2026
Classification	Internal / Confidential

1.2 Revision History

Version	Date	Author	Description
0.1	—	Joseph	Initial raw brainstorming / product-idea document (founder input).
1.0	21-Aug-2026	Joseph	First compiled Master PRD & SDLC document, structured from founder vision into a formal specification.

1.3 Purpose of This Document

This document is the single master reference for the AI Personalized Tourist Guide product (TC-SO1). It consolidates business requirements, functional and non-functional requirements, product design direction, technical architecture, AI strategy, testing strategy, and the full software development lifecycle into one coherent specification. It is written to be usable directly by a Product Manager, Business Analyst, UI/UX Designer, Developer, Solution Architect, QA Engineer, DevOps Engineer, and business stakeholder without needing a separate document per discipline.

Fields marked TBC (To Be Confirmed) indicate information that is genuinely not yet available and must be filled in during subsequent planning rather than assumed. Fields marked as a Proposed Target indicate a reasonable starting point suggested for this document, to be validated once real usage data exists.

2. Executive Summary

The AI Personalized Tourist Guide is a mobile-first AI travel companion that plans trips with the traveller, then stays with them for the entire journey — not just until a booking is confirmed. It combines conversational itinerary planning, real-time location-aware guidance, on-site heritage storytelling, basic travel-language assistance, and a private trip memory box into one integrated product, built around a single idea: the app should feel like travelling with a knowledgeable, personalised companion rather than operating a booking utility.

Most existing travel tools stop helping the moment a plan is finalised. Itinerary generators produce a list and disappear; OTAs (Online Travel Agencies) optimise for booking conversion, not the lived experience of the trip; and the richest layer of a destination — its history, culture and context — is still locked behind a paid human guide, which the founder's own experience shows can cost roughly ₹1,000 for a single visit to a single monument. Independently, India's linguistic diversity means this “understanding gap” is not only a problem for international travellers — a domestic traveller from Tamil Nadu visiting Delhi can face the same disorientation and language friction a foreign tourist would.

The product's primary users are individual and group leisure travellers, with an initial focus on domestic Indian travel where the combination of heritage depth, language diversity and guide cost creates the clearest opening. Its core loop is: describe your trip (interests, budget, time, location, and optionally your own existing plans) → receive and refine a personalised itinerary → travel with the app active as a location-aware companion that narrates points of interest, answers photo-based questions, helps with basic local phrases, and adapts the plan when something changes → return home with trip memories organised in a dedicated, trip-linked memory box.

What makes the product different is not any single feature — maps, translation, and AI chat all already exist as commodities — but the integration and the companion framing: one coherent experience that plans, guides, translates and remembers, in place of several disconnected point solutions and a paid human guide. This document defines that product in full: its business rationale, requirements, architecture, AI strategy, and delivery plan, structured so a lean or student-scale team can build a realistic MVP first and grow toward the full vision in phases.

3. Business Context

3.1 Background

Travel technology today is dominated by two categories of product. The first is booking-first platforms (OTAs and hotel/flight aggregators) whose core business is transaction conversion; personalisation in these products mostly means better search ranking, not a genuinely tailored plan. The second is itinerary-generation tools, including many recent AI chatbots, which produce a plan from a short prompt but rarely stay relevant once the traveller is actually on the ground, moving between locations in real time.

Separately, three long-standing point solutions exist outside the travel-app category altogether: paid local guides for on-site historical and cultural context, phrasebook or translation apps for basic communication, and generic photo-gallery apps for storing trip memories. Travellers today assemble their own experience from all of these tools manually. This product's thesis is that assembling them into one companion, around one evolving traveller profile, is itself the innovation — the same logic by which a modern smartphone combined many pre-existing technologies (camera, touchscreen, telephony) into one coherent product rather than inventing each piece from scratch.

3.2 Industry Context

India's domestic tourism base is large and growing, and it intersects with a specific structural feature of the country: enormous linguistic diversity across states. A traveller does not need to leave India to face a genuine language barrier — only to leave their home state. This is a distinct opportunity from typical “travel translation” products, which are usually designed around international/cross-border travel. (This is a qualitative observation, not a sized market estimate; no specific market-size figures are asserted in this document — see Section 39, Assumptions.)

3.3 Current Problems in Travel Planning

- Planning is fragmented across a search engine, a map app, a booking site, review sites, and personal notes, with no single tool holding the full picture.
- Personalisation in most itinerary tools is shallow — a handful of onboarding preferences that never evolve with actual behaviour.
- On-site historical and cultural context is either absent (a plaque with two lines of text) or expensive and inconsistent (a human guide, priced and quality-varying per visit).
- Language barriers are treated as an international-travel problem, leaving domestic travellers in linguistically different regions of their own country under-served.
- Trip memories (photos, notes) are scattered across a general camera roll with no trip-level structure, context, or easy way to revisit a specific journey.
- Group trips still rely on manual back-and-forth (chat threads, spreadsheets) to reconcile different people's budgets, interests and pace.

3.4 Opportunity

There is a clear opportunity to own the traveller relationship across the full trip lifecycle — before, during and after — rather than only at the moment of booking. A companion-style product that remains useful on the ground (not just during planning) has a natural reason to be opened repeatedly during a single trip, which is a stronger engagement pattern than a one-time itinerary generator. This also creates a credible long-term path toward monetisation through booking partnerships once trust and usage are established (see Section 31, Product Roadmap), without requiring booking revenue to justify building the MVP.

3.5 Why This Product Should Exist

No mainstream product currently combines personalised planning, real-time on-site companionship, grounded heritage storytelling, basic translation help, and trip memory-keeping into a single, coherent experience. Each capability exists somewhere; none of them exist together, wrapped in a single evolving relationship with the traveller. That combination — not any individual capability — is the product's reason to exist.

4. Problem Statement

4.1 Primary Problem

Travellers lack a single tool that plans a trip around their real preferences and then continues to actively help them during the trip itself — most tools either stop at planning or stop at booking, leaving the traveller unsupported for the part of the journey that matters most: the time actually spent at the destination.

4.2 Secondary Problems

- On-site cultural and historical understanding is inaccessible without hiring a paid human guide, and quality/cost vary widely visit to visit.
- Domestic travellers moving between linguistically different regions of India face real communication friction that is rarely addressed by travel apps built around international-travel assumptions.
- Trip memories accumulate with no trip-level organisation, making them hard to revisit, share, or preserve meaningfully.
- Group trips require manual, often frustrating reconciliation of different members' budgets, interests, and pace.
- Plans that were valid at creation time silently go stale — weather changes, an attraction closes, a delay happens — and most tools do not notice or adapt.

4.3 User Pain Points

- Decision fatigue during planning: too many open tabs, too many generic “top 10” lists that don't reflect personal interest.
- Disorientation on arrival: not knowing what's nearby, what's worth the time, or what's actually happening around the current location right now.
- **Guide cost as a real barrier:** a single guided visit to a major monument such as the Taj Mahal can cost roughly ₹1,000, which is a meaningful expense for a single stop on a single day.
- No natural way to ask a follow-up question about something specific (a carving, an inscription, an architectural detail) the way a live guide allows.
- Basic communication needs (asking directions, understanding a price) becoming a source of stress in an unfamiliar-language region.

4.4 Business Problems

- Existing travel products capture value at the booking moment and then largely lose the relationship, limiting repeat engagement and lifetime value.
- Generic personalisation (a short onboarding form) does not meaningfully differentiate one itinerary tool from another, making the category easy to commoditise.

4.5 Existing Gaps

No mainstream product currently unifies personalised planning, real-time on-site guidance, grounded heritage narration, basic translation assistance, and structured trip-memory keeping. Travellers substitute this gap with several disconnected tools plus, at times, a paid human guide.

4.6 Consequences of the Current Problem

- Travellers under-experience destinations relative to what they would understand with proper context.
- Money is spent on point solutions (guides, translation apps, separate planning tools) that a single well-designed companion could largely replace.

- Travel apps see a spike in usage during planning and then go unopened for the rest of the trip, weakening retention and the case for deeper personalisation.

5. Product Vision

5.1 Vision Statement

“To be the traveller's most trusted companion — before, during and after every journey — turning any place into a story worth understanding, in the traveller's own language and at their own pace.”

5.2 Mission

Build an AI companion that plans with the traveller rather than for them, stays actively useful once the trip begins, explains the meaning behind what the traveller is looking at, helps them communicate in unfamiliar-language regions, and holds on to their memories afterwards — replacing a stack of disconnected tools and a paid guide with one relationship that gets better the more it is used.

5.3 Product Philosophy

Personalisation is treated as a relationship that deepens over time, not a form filled in once at signup. The product should feel like travelling with a knowledgeable, attentive friend — present when useful, quiet when not — rather than operating a utility. Where the founder's own idea captured this as the app needing to know “the A-Z” of the person using it, this document treats that as a direction to grow into progressively (Section 20, Personalisation Engine) rather than a requirement to front-load at onboarding.

5.4 Long-Term Vision

In the long run, the product extends from a single-trip planner into an everyday companion for leisure time more broadly — weekend plans, outings with friends, and occasion-based local plans — built on the same personalisation engine as multi-day travel (Section 44 discusses how far to take this without diluting the core product). Once trust and usage are established, the product can also grow into a fuller travel transaction layer through booking partnerships, without that being a precondition for the MVP.

5.5 Value Proposition

One app that plans, guides, translates and remembers — in place of a separate itinerary tool, a paid on-site guide, a translation app, and a general photo gallery.

5.6 Unique Selling Proposition

The only travel app designed to act as an on-site AI companion — combining location-aware heritage storytelling and integrated basic translation — rather than stopping at itinerary generation.

5.7 Product Differentiation

Capability	Generic Itinerary App	OTA / Booking App	Static Audio Guide	This Product
Personalised planning	Basic, one-time	Minimal	None	Ongoing, evolves with use
Real-time on-site guidance	No	No	Partial (fixed script)	Yes — location-aware
Heritage / historical storytelling	No	No	Yes, but static & unaskable	Yes — plus follow-up Q&A
Language / translation help	No	No	No	Yes — phrasebook → live (phased)

Capability	Generic Itinerary App	OTA / Booking App	Static Audio Guide	This Product
Trip memory keeping	No	No	No	Yes — dedicated memory box
Primary business goal	Ad/lead gen	Booking conversion	One-time content sale	Ongoing companion relationship

6. Business Objectives

Business objectives for the product, distinct from the product-level objectives in Section 7:

- **Reduce planning effort:** lower the time and cognitive effort required to go from “I want to travel” to a usable, trustworthy plan. (Proposed Target / To Be Confirmed once baseline usage data exists.)
- **Increase personalisation depth:** move beyond onboarding-only personalisation toward recommendations that measurably improve as a user takes more trips.
- **Build habitual, companion-style engagement:** extend usage meaningfully beyond the booking moment, into the on-site and post-trip phases, to build a stronger retention pattern than one-time planning tools.
- **Establish a monetisation foundation:** lay the architectural and trust groundwork for future revenue via booking commissions and partner integrations, without requiring that revenue to validate the MVP.
- **Win a defensible early niche:** use India's domestic travel market — where heritage depth, guide cost, and regional language diversity intersect — as a differentiated wedge rather than competing head-on with large OTAs on booking price.

7. Product Objectives

What the application itself must achieve for a user, independent of business goals:

1. Generate a personalised itinerary from a combination of natural-language input, structured preferences (interests, budget, time, location), and any pre-existing trip ideas the user already has.
2. Track an active trip's location (with consent) and provide timely, relevant, location-aware recommendations and notifications.
3. Deliver grounded historical and cultural narration for a defined, growing set of supported points of interest.
4. Allow a user to ask a follow-up question about a specific place or detail using a photo.
5. Provide basic local-language phrase assistance for common travel needs, with live conversational translation as a later phase.
6. Let users capture and privately preserve trip memories (photos and notes) in a trip-linked space with a clear retention policy.
7. Support both solo and group/collaborative trip planning.
8. Continuously refine its understanding of a user's preferences from explicit feedback and observed behaviour, not only from onboarding.

8. Stakeholders

Stakeholder	Interest / Responsibility
Travellers (end users)	Primary users; want low-effort, genuinely relevant planning plus real, on-the-ground support during the trip.
Product Owner (Joseph)	Owns the product vision, prioritisation, and final scope decisions.
Product Manager	Owns the backlog, roadmap sequencing, and translating vision into shippable increments.
Business Analyst	Owns requirement clarity, traceability (Section 42), and stakeholder alignment.
UI/UX Designer	Owns the experience design across planning, on-trip, and post-trip modes (Section 25).
Frontend / Mobile Developers	Build the client experience across onboarding, planning, on-trip and memory-box flows.
Backend Developers	Build core APIs, business logic, auth, and integration orchestration (Section 26).
AI / ML Engineer	Owns personalisation, recommendation logic, and AI orchestration (LLM, RAG, guardrails — Section 19).
QA Engineer	Owns test strategy and quality gates across functional, AI-response, and non-functional testing (Section 34).
DevOps Engineer	Owns environments, CI/CD, deployment, and monitoring (Sections 35–36).
External API Providers	Maps, weather, translation, and LLM providers whose reliability and terms directly affect the product (Section 24).
Tourism / Heritage Bodies	Potential source of verified cultural and historical content, and future partnership opportunity (Section 22).
Booking Partners (Hotels/OTAs)	Future integration and monetisation partners once booking features are built (Section 24, Section 31).

9. Target Users & Personas

Five personas were selected because each represents a distinct planning pattern the product must support well; personas were not created for every conceivable traveller type, in line with the instruction to avoid unnecessary personas.

Meera, the Solo Culture-Seeker — 26, IT professional from Chennai, travelling alone to Delhi & Agra

Goals	Understand the history and significance of heritage sites in depth; navigate confidently despite not speaking Hindi.
Pain Points	Guide costs add up fast on a solo budget; generic listicles don't explain why a place matters; feels self-conscious asking basic questions repeatedly.
Needs	Grounded, on-demand heritage narration; basic phrase help; a way to feel safe travelling alone.
Behaviour	Researches heavily before a trip; budget-conscious; prefers doing things independently over joining group tours.
Expected Product Value	Replaces a paid guide and a translation app with one companion she already trusts with her plan.

The Weekend Squad — A group of 4–5 college friends planning a short trip together

Goals	Agree on a plan quickly without endless group-chat back-and-forth; keep the trip within a shared budget.
Pain Points	Reconciling different interests and budgets manually; someone always ends up doing the planning “work” alone.
Needs	Collaborative itinerary input; visible, fair handling of conflicting preferences; simple cost-splitting.
Behaviour	Plans on short notice; heavy chat/social app users; price-sensitive.
Expected Product Value	Turns group planning from a chore into a shared, low-friction activity.

The Family Traveller — Parents travelling with children and/or elderly parents

Goals	A comfortably paced itinerary that works for the whole family; reliable, low-surprise logistics.
Pain Points	Overpacked itineraries designed for solo/young travellers; too little buffer time; anxiety about the unexpected.
Needs	Pace-aware scheduling; accessibility-aware suggestions; dependable real-time updates.
Behaviour	Plans conservatively; values reliability over novelty; checks the plan repeatedly before and during the trip.
Expected Product Value	A plan that respects the group's actual pace, not a generic “must-see” list.

The Heritage Enthusiast — Domestic or NRI traveller with a deep interest in history and culture

Goals	Go beyond the surface — understand architecture, context, and stories most visitors miss.
Pain Points	Standard tours and plaques are too shallow; can't ask a specific question about a specific detail.
Needs	Deep, layered narration; the ability to point a camera at a detail and ask about it specifically.
Behaviour	Spends longer at fewer sites; actively seeks out lesser-known context.
Expected Product Value	The closest available substitute for an excellent private historian-guide, available on demand.

Arjun, the Local Planner — Uses the app for everyday plans, not only multi-day trips — a Saturday outing, a date, a catch-up with friends

Goals	A quick, personalised plan for a few hours based on mood, budget, time and location — without “trip planning” overhead.
Pain Points	Most travel apps are too heavyweight for a same-day local plan; generic “things to do nearby” lists ignore personal taste.
Needs	A lightweight planning mode reusing the same personalisation engine, scoped down for short local outings.
Behaviour	Opens the app between trips, not just when travelling — a key driver of everyday engagement.
Expected Product Value	Keeps the app relevant year-round, not only during travel season (see Section 44 for scope boundaries around this).

10. User Journey

10.1 Before the Trip

Discovery → Onboarding → Destination selection → Preference input (interests, budget, time, location, and optionally the user's own existing trip ideas) → AI itinerary generation → Conversational refinement → Booking (link-out in MVP; in-app in a later phase).

AI opportunity: every step from destination selection onward is a personalisation touchpoint — the assistant should ask clarifying questions the way a good travel agent would, rather than requiring a perfect prompt up front.

10.2 During the Trip

Location turned on → companion tracks position (with consent) → real-time recommendations and arrival notifications → on-site heritage narration at supported points of interest → photo-based Q&A for follow-up detail → translation/phrase assistance for communication → dynamic itinerary adaptation when weather, closures or delays occur → group sync, if travelling with others.

AI opportunity: this is the phase competitors are weakest in — the assistant should proactively surface only what's genuinely useful at that moment (arrival confirmations, a nearby highlight, a weather-driven adjustment) rather than pushing constant notifications.

10.3 After the Trip

Trip marked complete → memory box populated with photos and notes → trip summary/recap → review and rating prompt → feedback captured to refine future personalisation → preferences and learnings carried into the next trip.

AI opportunity: post-trip feedback (what was accepted, skipped, or corrected) is the highest-quality personalisation signal the product collects and should feed directly into the personalisation engine (Section 20).

11. Product Scope

11.1 In Scope

- AI-assisted itinerary generation from interests, budget, time, and location, including the ability to incorporate the user's own pre-existing trip ideas.
- Conversational chat interface for planning and iterative modification.
- Google (Gmail) OAuth sign-in.
- Google Maps integration for navigation and points of interest.
- Real-time, location-based recommendations and arrival notifications during an active trip.
- On-site heritage and historical narration for a curated, growing set of supported points of interest (starting deliberately small — see Section 29).
- Photo-based Q&A about a landmark or a specific detail of one.
- Basic local-language phrase assistance (MVP: curated phrasebook; live speech translation is a later phase — see Section 24 and Section 43).
- Trip memory box for photos and notes, with a defined retention policy and download capability.
- Trips list, collections, and favourites.
- User reviews and ratings based on past travellers' experience.
- A lightweight local/weekend outing planning mode, reusing the core personalisation engine.
- Group/collaborative trip planning (flagged as a Recommended Enhancement — see Section 12).

11.2 Out of Scope (Initial Release)

- In-app payment processing and automated flight/hotel booking — explicitly deferred by the product owner to a future phase; MVP links out to booking partners instead.
- Full live two-way speech-to-speech translation — technically demanding at launch; MVP ships text-based phrase assistance instead.
- Complete heritage-narration coverage of every monument nationwide — acknowledged as impractical; MVP covers a curated shortlist with a plan to expand (Section 29).
- Dating-specific matching or social-discovery features — the product supports occasion-based local outing planning, but is not a matchmaking platform (see Section 43 for the reasoning).
- Owning or managing third-party inventory (hotel/flight availability and pricing accuracy remain the responsibility of partner APIs).

12. Product Modules / Major Features

Modules are organised from the founder's raw ideas plus two Recommended Enhancements introduced by this analysis (Safety/SOS and Offline Heritage Access), each phase-tagged against the MVP definition in Section 29.

#	Module	Description	Phase
1	User Authentication	Google/Gmail OAuth sign-in.	MVP
2	Onboarding & Traveller Profile	Short interest/travel-style capture that seeds personalisation.	MVP
3	AI Conversational Planner	Chat-based planning, including free-form input of the user's own ideas.	MVP
4	Personalised Itinerary Engine	Generates and structures the day-by-day plan.	MVP
5	Real-Time Location Companion	Location tracking, arrival notifications, nearby recommendations.	MVP
6	Heritage & Cultural Narration	On-site storytelling for supported points of interest.	MVP (curated set)
7	Visual Q&A	Ask a question about a place using a photo.	MVP
8	Language & Translation Assistant	Text phrasebook (MVP) → live speech translation (later).	MVP / Phase 2
9	Maps & Navigation	Google Maps integration for routing and POI display.	MVP
10	Budget Planner & Tracker	Estimate at planning time; track actual spend on trip.	MVP / Phase 2
11	Attractions & Points of Interest	Curated + API-sourced places to recommend.	MVP
12	Restaurants / Food	Food recommendations aligned to preferences and budget.	MVP
13	Booking Integration	Link-out (MVP) → automated API booking (Future).	MVP / Future
14	Weather Intelligence	Forecast-aware planning and re-adaptation.	MVP
15	Trip Management	Trips list and trip status (upcoming/active/completed).	MVP
16	Collaborative Trip Planning	Multiple users contribute to one itinerary.	Phase 2
17	Memory Box	Trip-linked photo/notes storage with retention policy.	MVP
18	Collections & Favourites	Save and organise places of interest.	MVP
19	Reviews & Ratings	Past-traveller reviews on destinations/experiences.	MVP
20	Notifications	Arrival, disruption, and reminder notifications.	MVP
21	Weekend / Local Outing Planner	Lightweight, non-multi-day planning mode.	Phase 2
22	Safety & SOS	Trusted-contact location sharing for solo travellers.	Phase 2
23	Analytics (Internal)	Usage and quality signals for product improvement.	MVP

★RECOMMENDED ENHANCEMENT — SAFETY & SOS (MODULE 22)

Not present in the founder's original brainstorm. A trusted-contact location-sharing / SOS capability is proposed because the primary early persona (a solo traveller navigating an unfamiliar region) has a genuine safety need the current idea set doesn't address. User problem: solo travellers, especially in unfamiliar regions, have no quick way to share real-time location with someone they trust. Business value: safety is a strong trust and adoption driver for solo and family personas.

Technical complexity: low–medium (reuses the location infrastructure already required for Module 5). Recommended for Phase 2, immediately after MVP.

★RECOMMENDED ENHANCEMENT — OFFLINE HERITAGE ACCESS (SUPPORTS MODULE 6)

Not present in the founder's original brainstorm. Many heritage sites in India have patchy connectivity. Downloading narration content and maps for a trip in advance would let the on-site companion keep working where signal drops. User problem: the core heritage-narration feature is least reliable exactly where it matters most — inside old forts, monument complexes, and remote sites. Business value: protects the product's core differentiator from a foreseeable failure mode. Technical complexity: medium–high (requires a content sync and local-storage strategy). Recommended for Phase 2/3, evaluated further in Section 44.

13. Business Requirements

Business-oriented requirements, each with a stable ID for traceability through to functional requirements, user stories and test scenarios in Section 42.

ID	Requirement	Business Rationale	Priority	Stakeholder
BR-001	Generate a personalised itinerary from natural-language and structured inputs (interests, budget, time, location).	Core value proposition; the reason a user opens the app.	Must	Traveller, PM
BR-002	Authenticate users via Google (Gmail) OAuth.	Removes signup friction; avoids building/maintaining password auth.	Must	Traveller, Eng
BR-003	Provide real-time, location-aware recommendations and notifications during an active trip.	Differentiates from planning-only tools; drives on-trip engagement.	Must	Traveller, PM
BR-004	Provide narrated historical/cultural information for a defined set of supported points of interest.	Directly replaces the paid-guide cost pain point.	Must (curated set)	Traveller, Content
BR-005	Allow users to ask follow-up questions about a place via an uploaded/captured photo.	Extends narration value beyond a fixed script; matches how a live guide is actually used.	Should	Traveller, AI Eng
BR-006	Provide basic local-language phrase assistance for common travel needs.	Addresses domestic and cross-region language friction.	Should	Traveller
BR-007	Allow users to store trip photos/notes in a trip-linked memory space with a defined retention policy.	Creates a reason to return to the app after the trip; builds trust via a clear, honest data policy.	Must	Traveller
BR-008	Allow users to organise destinations into collections and mark favourites.	Supports pre-trip research and future trip planning.	Should	Traveller
BR-009	Support group trip planning where multiple users contribute preferences to one itinerary.	Addresses a real, currently manual pain point for group travel.	Should	Traveller, PM
BR-010	Allow users to submit and view reviews of destinations/experiences from past travellers.	Builds trust and content depth without the product authoring everything itself.	Should	Traveller
BR-011	Support lightweight, non-multi-day planning (weekend/local outings) using the same personalisation engine.	Drives usage between big trips; keeps the app relevant year-round.	Could	Traveller, PM
BR-012	Allow users to incorporate their own pre-existing trip ideas into the AI-generated itinerary.	Respects that many travellers already do some planning themselves; avoids ignoring user input.	Must	Traveller
BR-013	Architect the platform to support future automated hotel/flight booking via partner APIs.	Avoids a costly rebuild later; keeps the monetisation path open.	Could (design-for-future)	PM, Architect

ID	Requirement	Business Rationale	Priority	Stakeholder
BR-014	Protect user location and photo data with privacy-first controls and explicit consent.	Core trust requirement; regulatory and reputational risk if absent.	Must	All
BR-015	Provide a safety mechanism (trusted-contact location sharing / SOS) for solo travellers.	Addresses a real safety need for the primary early persona (Recommended Enhancement).	Should	Traveller
BR-016	Capture user feedback after each trip to refine future personalisation.	Highest-quality personalisation signal available to the product.	Should	Traveller, AI Eng
BR-017	Dynamically re-evaluate and propose itinerary adjustments in response to disruptions (weather, closures, delays).	Keeps the plan trustworthy once conditions change, not just at creation time.	Should	Traveller, AI Eng
BR-018	Allow users to track actual spending against their planned budget during a trip.	Extends budget planning from an estimate into a genuinely useful on-trip tool.	Could	Traveller
BR-019	Capture user interests and traveller type during onboarding to seed initial personalisation.	Solves the cold-start problem for a brand-new user.	Must	Traveller, AI Eng

14. Functional Requirements

Each functional requirement below is written to be specific and testable, avoiding vague statements such as “the system should provide a good itinerary.” IDs trace back to Section 13 (Business Requirements) and forward to Section 17 (User Stories) and Section 42 (Traceability Matrix).

FR-001 — Generate Personalised Itinerary

Purpose	Produce a day-by-day itinerary matching the traveller's stated interests, budget, time window and location.
Actor	Traveller
Preconditions	User is authenticated (or in guest mode, if permitted).
Trigger	User submits trip parameters via chat or a structured form.
Main Flow	1. User provides interests, budget, time window, and destination (chat and/or form). 2. System merges these with the onboarding profile (FR-003) and any pasted trip ideas (FR-005). 3. AI planning layer generates a candidate day-by-day itinerary. 4. System validates the itinerary against business rules (Section 16) — budget, opening hours, pacing. 5. Itinerary is presented with an option to refine conversationally (FR-004).
Alternative Flow	If some inputs are missing, the system asks up to 2–3 targeted clarifying questions instead of failing outright.
Exception Flow	If the AI planning layer fails or times out, the system shows a curated fallback itinerary template for the destination and flags that live generation failed.
Inputs	Interests, budget, time window, destination, onboarding profile, optional free-text trip ideas.
Outputs	Structured day-by-day itinerary (POIs, suggested times, estimated cost).
Business Rules	Total estimated cost must not exceed stated budget by more than a Proposed 10% tolerance without explicit user override (see Section 16).
Acceptance Criteria	Given valid interests, budget, time and location, when the user submits the request, then an itinerary is returned within the target response time (Section 15). Given an itinerary exceeding budget by more than the tolerance, when it is generated, then the system flags the overage before presenting it.
Priority	Must

FR-002 — Google OAuth Sign-In

Purpose	Authenticate users with minimal friction using their existing Google account.
Actor	Traveller
Preconditions	User has a Google account.
Trigger	User taps “Sign in with Google”.
Main Flow	1. User taps Google sign-in. 2. App redirects to Google's OAuth consent screen. 3. User approves requested scopes. 4. App receives and validates the auth token, creates/loads the user's profile.
Alternative Flow	Returning user with a valid session is signed in silently without repeating the OAuth flow.
Exception Flow	If the user denies consent or the OAuth call fails, the app shows a clear retry option and does not create a partial account.

Inputs	Google OAuth token/consent.
Outputs	Authenticated session; user profile created or loaded.
Business Rules	No passwords are stored by the platform for MVP; Google is the sole identity provider.
Acceptance Criteria	Given a valid Google account, when the user completes consent, then they land on the home/dashboard screen signed in. Given consent is denied, when the user returns to the app, then no account is created and the sign-in screen reappears.
Priority	Must

FR-003 — Onboarding Interest & Traveller-Type Capture

Purpose	Seed initial personalisation with a short set of preference questions right after first sign-in.
Actor	New traveller
Preconditions	User has just completed sign-in for the first time.
Trigger	First successful login.
Main Flow	1. App presents a short set of interest/travel-style questions (e.g. pace, budget bracket, interest themes). 2. User answers (single/multi-select where possible to minimise typing). 3. Responses are saved to the traveller profile. 4. User proceeds to the home/dashboard screen.
Alternative Flow	User may skip onboarding; the system falls back to popular/curated defaults until enough behavioural data exists.
Exception Flow	If saving the profile fails, the user is not blocked from proceeding — defaults are used and a retry happens silently in the background.
Inputs	Selected interests, travel style, pace, rough budget bracket.
Outputs	Initial traveller profile record.
Business Rules	Onboarding must be completable in a small number of steps (Proposed Target: under 6 screens) to protect completion rate (Section 15, Usability).
Acceptance Criteria	Given a first-time user, when they complete onboarding, then their first itinerary reflects at least one stated interest. Given a user skips onboarding, when they request an itinerary, then curated defaults are used without error.
Priority	Must

FR-004 — Conversational Itinerary Modification

Purpose	Let users refine a generated itinerary through natural conversation instead of manual form editing.
Actor	Traveller
Preconditions	An itinerary has already been generated (FR-001).
Trigger	User sends a chat message requesting a change (e.g. “swap day 2 for something more relaxed”).
Main Flow	1. User sends a modification request in chat. 2. System interprets the request against the current itinerary and profile. 3. System proposes an updated itinerary segment. 4. User accepts, further refines, or rejects the change.

Alternative Flow	If the request is ambiguous, the system asks a single clarifying question rather than guessing silently.
Exception Flow	If a requested change violates a hard business rule (e.g. pushes an activity outside opening hours), the system explains why and offers the closest valid alternative.
Inputs	Free-text chat message, current itinerary state.
Outputs	Updated itinerary segment, chat response.
Business Rules	An explicit user correction always overrides an AI-inferred preference for the remainder of the session and future recommendations (Section 16).
Acceptance Criteria	Given an existing itinerary, when the user requests a specific change, then only the relevant segment changes — the rest of the plan is preserved. Given an invalid requested change, when submitted, then the system explains the conflict and proposes an alternative instead of silently ignoring the request.
Priority	Must

FR-005 — Incorporate User-Provided Trip Ideas

Purpose	Let users paste or type their own existing trip ideas/notes so the AI builds on them instead of starting from a blank plan.
Actor	Traveller
Preconditions	User is creating or editing a trip.
Trigger	User pastes/types free-text notes into a “my own ideas” input during trip creation.
Main Flow	1. User enters free-text notes (places, dates, must-dos) alongside or instead of structured fields. 2. System extracts identifiable structured elements (places, dates, activities) from the text. 3. Extracted elements are merged into the itinerary generation input for FR-001. 4. Any unrecognised parts of the note are preserved and shown alongside the generated plan for user reference.
Alternative Flow	If nothing structured can be extracted, the system still passes the raw notes to the AI planner as context rather than discarding them.
Exception Flow	If extraction produces a conflicting or unrealistic combination (e.g. two far-apart cities on the same day), the system flags the conflict rather than silently resolving it.
Inputs	Free-text user notes.
Outputs	Structured elements merged into the itinerary; unrecognised notes retained for reference.
Business Rules	User-provided ideas are never silently dropped — anything not machine-parsable is still shown back to the user.
Acceptance Criteria	Given a user pastes a paragraph naming 2–3 specific places, when the itinerary is generated, then those places appear in the plan. Given notes the system cannot parse, when the plan is generated, then the original notes remain visible to the user.
Priority	Must

FR-006 — Real-Time Location Tracking & Arrival Notification

Purpose	Track an active trip's location (with consent) to enable on-site recommendations and confirm arrivals.
Actor	Traveller (on an active trip)
Preconditions	Trip status is “active”; user has granted location permission.

Trigger	User enters the geofence radius of a planned itinerary stop, or opens the app during an active trip.
Main Flow	1. App requests/confirms location permission at trip start. 2. Device location is checked against the active itinerary's planned stops. 3. On entering a stop's geofence, the system sends an arrival notification. 4. Nearby, relevant recommendations are surfaced based on current position.
Alternative Flow	If the user is off-route from the plan, the system offers to re-centre suggestions around actual current location instead of the original plan.
Exception Flow	If location permission is denied or unavailable, location-dependent features are disabled gracefully; the rest of the app remains usable (manual “I'm at” input as fallback).
Inputs	Device GPS location, active itinerary.
Outputs	Arrival notifications, nearby recommendations.
Business Rules	Location is tracked only while a trip is active and only with explicit consent; see Section 27 for retention limits.
Acceptance Criteria	Given location permission is granted and a trip is active, when the user enters a planned stop's radius, then an arrival notification is sent within the target latency (Section 15). Given location permission is denied, when the user opens the app, then core planning features still function normally.
Priority	Must

FR-007 — On-Site Heritage Narration Playback

Purpose	Deliver grounded historical/cultural narration for a supported point of interest when the traveller is on-site.
Actor	Traveller (on-site at a supported POI)
Preconditions	POI is in the supported narration catalog; user is at or near the location.
Trigger	User opens the POI's narration screen (manually, or via a proactive on-arrival prompt).
Main Flow	1. System identifies the POI from location or user selection. 2. System retrieves verified reference content for that POI (RAG-grounded, Section 19). 3. Narration is presented in layers — a short overview plus deeper detail on request (Section 22). 4. User can navigate between sub-sections (e.g. entrance, main structure, specific wings) as they physically move.
Alternative Flow	If only partial content exists for a POI, the system presents what is verified and clearly marks the rest as unavailable rather than generating unverified filler.
Exception Flow	If the POI is not in the supported catalog, the system says so plainly and offers to log it as a request for future coverage, instead of improvising unverified narration.
Inputs	POI identifier/location, curated reference content.
Outputs	Layered narration content (text, optionally audio).
Business Rules	Narration must only state claims the system can support from curated/verified sources; uncertain claims are flagged, never presented as fact (Section 16, Section 19).
Acceptance Criteria	Given a supported POI, when the user opens narration, then content grounded in verified sources is shown within the target load time. Given an unsupported POI, when selected, then the system clearly states it is not yet covered rather than fabricating content.
Priority	Must (curated set)

FR-008 — Photo-Based Landmark Q&A

Purpose	Let a user photograph a place or a specific detail and ask a follow-up question about it.
Actor	Traveller (on-site)
Preconditions	User is viewing or has access to the narration/companion screen.
Trigger	User captures or uploads a photo and asks a question.
Main Flow	1. User taps the camera/photo-question icon. 2. User captures or selects a photo and types or speaks a question. 3. System analyses the photo alongside the current POI context. 4. System returns an answer grounded in available reference content, or an honest “not confident” response.
Alternative Flow	If the photo cannot be matched to the current POI's known content, the system answers as generally as it confidently can and invites the user to rephrase or specify more.
Exception Flow	If image analysis fails (poor quality, unsupported format), the system asks for a retake rather than guessing from an unusable image.
Inputs	Photo, free-text/voice question, current location/POI context.
Outputs	Grounded answer text.
Business Rules	Low-confidence answers must be visibly flagged as such rather than presented with the same certainty as verified narration (Section 19).
Acceptance Criteria	Given a photo of a recognisable detail at a supported POI, when a question is asked, then a relevant, grounded answer is returned. Given a low-confidence match, when answered, then the response is visibly flagged as uncertain.
Priority	Should

FR-009 — Local Phrase Assistant

Purpose	Provide quick access to common travel phrases in the local language of the current destination.
Actor	Traveller
Preconditions	Destination/region is known (from active trip or manual selection).
Trigger	User opens the phrase assistant, or the system proactively surfaces it on arrival in a new linguistic region.
Main Flow	1. System determines the relevant local language from the destination. 2. User browses or searches common categories (directions, prices, essentials, courtesy phrases). 3. Phrase is shown in the local script plus a phonetic transcription. 4. (Phase 2) User can speak/type in their own language for live translation — see Section 24, Section 43.
Alternative Flow	If the destination spans multiple locally-relevant languages, the system lets the user pick which one to display.
Exception Flow	If the phrasebook has no content for a given region yet, the system states this clearly rather than showing an empty or broken screen.
Inputs	Destination/region, phrase category selection.
Outputs	Phrase text, transliteration, (later) audio.
Business Rules	MVP phrase content must work without a live network connection once downloaded for a trip (ties to the offline enhancement, Section 12).

Acceptance Criteria	Given a destination with phrasebook coverage, when the user opens the assistant, then relevant common phrases are shown with transliteration. Given no network connection but a pre-downloaded trip, when opened, then phrases still load.
Priority	Should

FR-010 — Trip Memory Box Upload & Storage

Purpose	Let users store trip photos and notes in a private, trip-linked space.
Actor	Traveller
Preconditions	A trip exists (active or completed).
Trigger	User uploads a photo/note to a specific trip's memory box.
Main Flow	1. User opens a trip and selects “Add to Memory Box”. 2. User uploads photo(s) and/or a note. 3. System stores the content, linked to that trip, in encrypted storage. 4. Content appears in the trip's memory box gallery, viewable any time.
Alternative Flow	User can download individual items or the full memory box at any time rather than only before expiry.
Exception Flow	If upload fails (network/size), the system retries or clearly reports failure without silently dropping the file.
Inputs	Photo/video files, optional notes, trip ID.
Outputs	Stored, trip-linked memory items; downloadable exports.
Business Rules	Default retention is 12 months from upload, with a reminder before expiry and a one-tap download/extend action — never silent deletion (Section 39, Section 43).
Acceptance Criteria	Given a completed upload, when the user reopens the trip, then the item appears in that trip's memory box. Given an item approaching its retention limit, when the limit nears, then the user receives a reminder before any deletion.
Priority	Must

FR-011 — Create Collection / Mark Favourite

Purpose	Let users save and organise places of interest for current or future trip planning.
Actor	Traveller
Preconditions	User is viewing a place/POI.
Trigger	User taps “favourite” or “add to collection”.
Main Flow	1. User taps favourite on a POI, or creates/selects a named collection. 2. System saves the association to the user's profile. 3. Saved items are retrievable from the Collections/Favourites screen and can inform future itinerary generation.
Alternative Flow	User can create multiple named collections (e.g. “Food to try”, “Next Kerala trip”).
Exception Flow	None beyond standard save-failure retry handling.
Inputs	POI reference, optional collection name.
Outputs	Saved favourite/collection entries.
Business Rules	Favourited items are a personalisation signal (Section 20), not just a static list.

Acceptance Criteria	Given a POI, when the user favourites it, then it appears under Favourites immediately. Given a named collection, when a POI is added, then it appears only in that collection, not all collections.
Priority	Should

FR-012 — Group Trip Collaborative Planning

Purpose	Let multiple users contribute preferences to a single shared itinerary.
Actor	Trip organiser and invited members
Preconditions	A trip exists; organiser has invited at least one other member.
Trigger	Organiser invites members, or a member submits their preferences to a shared trip.
Main Flow	1. Organiser creates a trip and invites members (link or in-app invite). 2. Each member submits their own interests/budget/constraints. 3. System generates a single itinerary reconciling inputs, flagging any conflicts. 4. Members view the shared plan and can propose changes visible to the group.
Alternative Flow	If members' budgets differ significantly, the system proposes budget-tiered alternatives for shared activities rather than forcing one number.
Exception Flow	If a member has not responded by a set point, the organiser can proceed with available inputs; the system marks which members' preferences were included.
Inputs	Multiple members' preferences, trip ID.
Outputs	Single reconciled itinerary; conflict flags.
Business Rules	Where preferences conflict, the system proposes a reconciled option and explains the trade-off rather than silently choosing one member's preference (Section 16).
Acceptance Criteria	Given 2+ members with differing interests, when the itinerary is generated, then conflicting preferences are visibly flagged with the resolution shown. Given a non-responding member, when the organiser proceeds, then the plan clearly shows whose preferences were included.
Priority	Should

FR-013 — Submit / View Reviews

Purpose	Let travellers read reviews from past users and submit their own after a visit.
Actor	Traveller
Preconditions	For submission: user has a completed trip that included the place.
Trigger	User opens a place's detail screen (view), or completes a trip (submit prompt).
Main Flow	1. User views a place and sees aggregated past-traveller reviews. 2. After a completed trip, the user is prompted to rate/review visited places. 3. Submitted reviews are moderated for basic policy compliance before becoming visible. 4. Reviews factor into future recommendation ranking (Section 19).
Alternative Flow	User may skip reviewing; the app does not block trip completion on it.
Exception Flow	Reviews flagged by moderation are hidden pending review rather than published automatically.
Inputs	Rating, review text, place/trip reference.
Outputs	Published review; updated place rating aggregate.

Business Rules	Only users who completed a trip including that place may review it, to preserve trust in the review content.
Acceptance Criteria	Given a completed trip, when the user is prompted, then they can rate and review each visited place in under a minute. Given a submitted review, when it passes moderation, then it appears on the place's detail screen.
Priority	Should

FR-014 — Dynamic Itinerary Re-Adaptation

Purpose	Re-evaluate and propose itinerary adjustments when conditions change during an active trip.
Actor	Traveller (on active trip)
Preconditions	Trip is active; a monitored disruption trigger occurs.
Trigger	Weather change, POI closure, delay, missed activity, or explicit user request to adjust.
Main Flow	1. System detects a relevant disruption (or receives a user request). 2. Remaining itinerary is re-evaluated against current conditions. 3. System proposes a specific adjustment (swap, reorder, or skip) with a short reason. 4. User confirms, edits, or dismisses the proposal — nothing changes automatically without confirmation.
Alternative Flow	If multiple valid adjustments exist, the system offers the top 2–3 options rather than a single forced choice.
Exception Flow	If no reasonable alternative exists (e.g. everything nearby is also closed), the system says so plainly rather than forcing a poor suggestion.
Inputs	Live weather/closure/delay signals, current itinerary state.
Outputs	Proposed itinerary adjustment(s).
Business Rules	The system never silently alters a confirmed plan — every change is presented for confirmation first (Section 16, Section 21).
Acceptance Criteria	Given a monitored POI closes unexpectedly, when detected, then the user receives an adjustment proposal before the scheduled visit time. Given a proposed adjustment, when the user dismisses it, then the original plan remains unchanged.
Priority	Should

FR-015 — Weekend / Local Outing Quick Plan

Purpose	Generate a short, lightweight plan for a same-day or single-day local outing, distinct from full multi-day trip planning.
Actor	Traveller (local user)
Preconditions	None beyond an authenticated session.
Trigger	User selects “Quick Plan” / “Local Outing” instead of “Plan a Trip”.
Main Flow	1. User specifies time available, budget, and mood/occasion (e.g. “casual”, “date”, “with friends”). 2. System generates a short plan (1–3 stops) using the same personalisation engine as full trips, scoped down. 3. User can start the plan immediately with live location-based guidance (reusing FR-006), without full trip/heritage-narration overhead.
Alternative Flow	User can convert a quick plan into a saved favourite/collection for reuse.

Exception Flow	If insufficient local data exists for the area, the system says so rather than returning a low-quality generic list.
Inputs	Time available, budget, occasion/mood, current location.
Outputs	Short local outing plan.
Business Rules	Quick Plans intentionally exclude multi-day logistics and heritage narration to stay lightweight (Section 43).
Acceptance Criteria	Given time, budget and occasion, when submitted, then a plan with 1–3 relevant stops is returned quickly. Given a generated quick plan, when the user starts it, then arrival notifications work the same way as for full trips.
Priority	Could

FR-016 — Budget Tracking Against Itinerary

Purpose	Let users log actual spend during a trip and compare it against the planned budget.
Actor	Traveller (on active trip)
Preconditions	An itinerary with an estimated budget exists.
Trigger	User logs an expense, or links a booking confirmation with a known cost.
Main Flow	1. User logs an expense against a trip (manual entry, or auto-captured from an in-app booking in later phases). 2. System updates running actual-vs-planned totals. 3. If actual spend trends over the planned budget, the system surfaces a proactive notice.
Alternative Flow	For group trips, expenses can optionally be split and tracked per member (ties to Section 44).
Exception Flow	None beyond standard input-validation handling for malformed amounts.
Inputs	Expense amount, category, trip ID.
Outputs	Updated spend total; over-budget notice if applicable.
Business Rules	Budget tracking is informational in MVP — it does not block or auto-cancel any plan item.
Acceptance Criteria	Given a logged expense, when saved, then the trip's actual-spend total updates immediately. Given actual spend nearing the planned budget, when the threshold is crossed, then the user is notified.
Priority	Could

FR-017 — Safety / SOS Trusted-Contact Sharing

Purpose	Let a solo (or any) traveller share live location with a trusted contact and access a quick SOS action.
Actor	Traveller
Preconditions	User has granted location permission and added at least one trusted contact.
Trigger	User enables “Share my trip”, or taps the SOS control.
Main Flow	1. User adds a trusted contact (name + phone/email). 2. User enables live location sharing for the active trip. 3. Trusted contact receives a link to view live location for the duration of the trip. 4. If SOS is triggered, the trusted contact is immediately notified with the user's last known location.
Alternative Flow	User can stop sharing at any time; sharing auto-expires when the trip ends.

Exception Flow	If location is unavailable when SOS is triggered, the last known location and timestamp are sent instead of failing silently.
Inputs	Trusted contact details, location stream, SOS trigger.
Outputs	Shared-location link; SOS alert to trusted contact.
Business Rules	Location sharing is strictly opt-in per trip and never enabled by default (Section 27).
Acceptance Criteria	Given sharing is enabled, when the trusted contact opens the link, then they see current live location for that trip only. Given SOS is triggered, when sent, then the trusted contact is notified within the target latency (Section 15).
Priority	Should (Recommended Enhancement)

FR-018 — Post-Trip Feedback Capture

Purpose	Capture a short, private feedback signal after trip completion to improve future personalisation — distinct from the public review in FR-013.
Actor	Traveller
Preconditions	Trip status changes to “completed”.
Trigger	Trip marked complete (automatically at end date, or manually by the user).
Main Flow	1. System prompts a short private feedback flow (e.g. what worked, what didn't, per-stop thumbs up/down). 2. User responds (fully, partially, or skips). 3. Signals are written to the traveller's personalisation profile (Section 20).
Alternative Flow	If the user skips, the system relies on implicit signals only (accepted/rejected suggestions during the trip).
Exception Flow	None beyond standard save-retry handling.
Inputs	Per-stop and overall trip feedback.
Outputs	Updated personalisation profile signals.
Business Rules	An explicit negative signal on a specific recommendation must suppress similar future suggestions, not just log silently (Section 16).
Acceptance Criteria	Given a completed trip, when the user provides feedback, then their next itinerary reflects at least one piece of that feedback. Given the user skips feedback, when the next itinerary is generated, then generation still proceeds normally using implicit signals.
Priority	Should

15. Non-Functional Requirements

Targets marked “Proposed Target / TBC” are reasonable starting points suggested for planning purposes; they are not committed SLAs and should be validated once real usage data exists.

Category	Requirement	Target
Performance	Itinerary generation and chat responses should feel conversational, not batch-like.	Proposed Target: itinerary generation within 8–12s; chat replies within 3–5s.
Scalability	The API layer should scale horizontally independent of the AI/recommendation layer.	Concurrent-user target: TBC based on launch scope.
Availability	Core planning and on-trip companion services should be reliably available, especially during active trips.	Proposed Target: 99.5% uptime for core services.
Reliability	The system should degrade gracefully, not fail outright, when a dependency (LLM, Maps, Weather) is unavailable.	Fallback to cached/rule-based suggestions; no hard crash on dependency failure.
Security	Data in transit and at rest must be encrypted; access must be least-privilege.	TLS in transit; encryption at rest for location history and photos.
Privacy	Location and photo data require explicit, granular consent and minimal retention.	See Section 27 for full detail.
Usability	Onboarding and core flows should be quick to complete without prior training.	Proposed Target: onboarding completable in under 6 screens.
Accessibility	Core screens should be usable by people relying on assistive technology.	Proposed Target / TBC: WCAG 2.1 AA aspiration for MVP screens.
Maintainability	Services should have clear module boundaries and documented APIs.	API documentation required for every service boundary before release.
Observability	Engineers should be able to trace a request end-to-end and see system health at a glance.	Centralised logging, tracing and alerting (Section 36).
Data Integrity	Itinerary, booking and profile data must remain consistent under concurrent updates.	Input validation and idempotent write operations on all mutating endpoints.
Disaster Recovery	The system must be recoverable after data loss or an infrastructure incident.	Proposed Target / TBC: RPO/RTO to be defined; automated daily backups minimum.

16. Business Rules

- **Budget:** total estimated itinerary cost must not exceed the stated budget by more than a Proposed 10% tolerance without explicit user override.
- **Travel Time / Dates:** itinerary items must not overlap, and travel buffer time between geographically distant stops must be accounted for.
- **Locations:** recommendations are filtered to what is realistically reachable within the trip's available time.
- **Opening Hours:** the system must not schedule a visit outside a POI's known operating hours; if hours are unknown, the item is marked "verify on arrival."
- **Activity Duration:** every itinerary item carries an estimated duration, used for daily pacing calculations.
- **Travel Distance:** consecutive stops are sequenced to minimise backtracking (basic route optimisation).
- **User Preferences:** an explicit user preference always overrides an AI-inferred preference when the two conflict.
- **Group Preferences:** when group members' preferences conflict, the system proposes a reconciled option and visibly flags the trade-off rather than silently picking one member's preference.
- **Booking Availability:** the system never presents a booking option as available without verifying against the source, or it is explicitly labelled "unverified."
- **Weather:** outdoor activities affected by adverse forecast weather are flagged with a suggested indoor or alternative option.
- **Personalisation:** an explicit user correction (e.g. "I don't like this") is retained and applied to future recommendations, not only the current session.
- **AI Recommendations:** AI-generated historical or cultural narration is limited to claims the system can support with reasonable confidence; uncertain claims are flagged rather than presented as fact.

17. User Stories

Only stories that map to a distinct, meaningful behaviour are included — each traces to a functional requirement in Section 14 and into the traceability matrix in Section 42.

Epic: Planning

US-001 As a traveller, I want to generate a personalised itinerary by specifying my interests, budget, time and location, so that I get a relevant plan without manual research.

- **Given/When/Then:** *Given valid trip parameters, when I submit them, then I receive a structured itinerary matching my budget and time window.*

US-002 As a traveller, I want to paste or type my own trip ideas, so that the AI builds on what I already planned instead of ignoring it.

- **Given/When/Then:** *Given I paste notes naming specific places, when my itinerary is generated, then those places appear in the plan.*

US-003 As a traveller, I want to refine my itinerary conversationally, so that I can adjust it without starting over.

- **Given/When/Then:** *Given an existing itinerary, when I ask for a specific change in chat, then only the relevant part of the plan updates.*

Epic: Onboarding

US-004 As a new user, I want to answer a short set of questions about my interests and travel style, so that my first itinerary already feels personalised.

- **Given/When/Then:** *Given I complete onboarding, when my first itinerary is generated, then it reflects at least one interest I selected.*

Epic: On-Trip Companion

US-005 As a traveller, I want to be notified when I arrive at a planned location, so that I know my navigation worked and what to do next.

- **Given/When/Then:** *Given an active trip and location access, when I enter a planned stop's radius, then I receive an arrival notification.*

US-006 As a traveller, I want to get location-aware suggestions near my current position, so that I can discover relevant things nearby without searching manually.

- **Given/When/Then:** *Given an active trip, when I open the app near a point of interest, then nearby relevant suggestions are shown.*

US-007 As a traveller, I want to have my itinerary adjust when something changes (weather, closures), so that I don't have to replan from scratch.

- **Given/When/Then:** *Given a monitored disruption occurs, when detected, then I receive a proposed adjustment before the affected activity's scheduled time.*

Epic: Heritage & Language

US-008 As a solo traveller visiting an unfamiliar region, I want to get on-site historical narration for major landmarks, so that I understand what I'm seeing without paying for a human guide.

- **Given/When/Then:** *Given I am at a supported landmark, when I open narration, then I receive grounded historical content within the target load time.*

US-009 As a traveller, I want to photograph a part of a monument and ask a question about it, so that I can get detail beyond the general narration.

- **Given/When/Then:** *Given a photo of a recognisable detail, when I ask a question, then I receive a grounded, relevant answer.*

US-010 As a traveller in a region where I don't speak the local language, I want to get quick access to common phrases (directions, prices), so that I can communicate basic needs confidently.

- **Given/When/Then:** *Given a destination with phrasebook coverage, when I open the assistant, then relevant phrases appear with transliteration.*

Epic: Memory Box

US-011 As a traveller, I want to upload photos to a trip-specific memory space, so that my trip memories stay organised and don't get lost in my camera roll.

- **Given/When/Then:** *Given a completed upload, when I reopen the trip, then the photo appears in that trip's memory box.*

US-012 As a traveller, I want to download my trip photos before they expire from the memory box, so that I don't lose them permanently.

- **Given/When/Then:** *Given an item nearing its retention limit, when the limit approaches, then I am reminded with a one-tap download option.*

Epic: Group & Local Planning

US-013 As a group of friends planning a trip together, I want to combine our individual preferences into one itinerary, so that we don't have to negotiate everything manually.

- **Given/When/Then:** *Given 2+ members submit preferences, when the itinerary is generated, then it reflects all members' input.*

US-014 As a group member, I want to see where our preferences conflict and what was chosen instead, so that planning feels fair and transparent.

- **Given/When/Then:** *Given conflicting group preferences, when the itinerary is generated, then the conflict and its resolution are visibly flagged.*

US-015 As a user planning a weekend outing, not a multi-day trip, I want to get a quick plan based on time, budget and mood, so that I can use the app for everyday plans, not just big trips.

- **Given/When/Then:** *Given time, budget and occasion, when I request a Quick Plan, then I receive a short plan with 1–3 relevant stops.*

Epic: Trust & Safety

US-016 As a solo traveller, I want to share my live location with a trusted contact, so that someone knows where I am during my trip.

- **Given/When/Then:** *Given sharing is enabled, when my trusted contact opens the link, then they see my current live location for that trip only.*

Epic: Reviews & Feedback

US-017 As a traveller choosing a destination, I want to read reviews from past travellers, so that I can set realistic expectations.

- **Given/When/Then:** *Given a place with past visits, when I view its detail screen, then I see aggregated reviews from travellers who actually went.*

US-018 As a traveller who just finished a trip, I want to rate my experience, so that future recommendations for me (and others) improve.

- **Given/When/Then:** *Given a completed trip, when I submit feedback, then my next itinerary reflects at least one piece of that feedback.*

Epic: Trip & Collections Management

US-019 As a traveller, I want to save destinations to a favourites list or collection, so that I can plan future trips around things I've already found interesting.

- **Given/When/Then:** *Given a place I like, when I favourite it, then it appears under Favourites immediately.*

US-020 As a traveller, I want to see all my past and upcoming trips in one place, so that I have a single view of my travel history.

- **Given/When/Then:** *Given I have multiple trips, when I open Trips, then I see them grouped by upcoming, active and completed.*

18. Use Cases

Six use cases cover the product's primary, distinguishing interactions in full flow detail. Narrower interactions are already captured at the functional-requirement level in Section 14.

UC-01 — Generate a Personalised Itinerary

Actor	Traveller
Goal	Obtain a day-by-day itinerary matching stated preferences.
Preconditions	User is signed in.
Trigger	User starts “Plan a Trip.”
Main Flow	1. User enters destination, dates, budget and interests (and optionally own trip ideas). 2. System generates a candidate itinerary. 3. System validates it against business rules (Section 16). 4. User reviews the itinerary and either accepts or requests changes (UC-02 equivalent handled via FR-004).
Alternative Flow	User asks the assistant to “surprise me” with minimal input; system uses onboarding profile + popular defaults.
Exception Flow	AI generation fails → curated fallback template shown, failure logged.
Postconditions	A draft itinerary exists, linked to the user's account, ready for booking or further refinement.

UC-02 — Real-Time On-Site Companion Guidance

Actor	Traveller (on active trip)
Goal	Receive timely, relevant guidance while physically at the destination.
Preconditions	Trip is active; location permission granted.
Trigger	User's location changes during an active trip.
Main Flow	1. System continuously (or periodically) checks location against the itinerary. 2. On entering a planned stop's radius, an arrival notification is sent. 3. Nearby relevant recommendations are surfaced. 4. If a disruption is detected, the system proposes an adjustment (FR-014).
Alternative Flow	User is off-route → system re-centres suggestions on actual location rather than the original plan.
Exception Flow	Location unavailable → location-dependent features disabled gracefully; manual “I'm at” fallback offered.
Postconditions	User has received relevant, timely, on-the-ground support without manually searching for it.

UC-03 — On-Site Heritage Narration & Photo Q&A

Actor	Traveller (on-site at a supported POI)
Goal	Understand the historical/cultural significance of a place, including specific details, without a paid guide.
Preconditions	POI is in the supported narration catalog.
Trigger	User opens narration for a POI, or asks a photo-based question.
Main Flow	1. User opens the POI's narration screen (or it is proactively suggested on arrival).

	2. System presents layered narration — overview, then deeper detail on request. 3. User optionally photographs a specific detail and asks a follow-up question. 4. System answers, grounded in verified content, or clearly states low confidence.
Alternative Flow	User navigates narration sub-sections as they physically move through the site.
Exception Flow	POI not yet supported → system states this plainly and logs the request rather than fabricating content.
Postconditions	User leaves with a substantially deeper understanding of the site than a plaque alone would provide.

UC-04 — Group Trip Collaborative Planning

Actor	Trip organiser and invited members
Goal	Produce one itinerary that fairly reflects multiple people's preferences.
Preconditions	Organiser has created a trip and invited members.
Trigger	A member submits preferences to a shared trip.
Main Flow	1. Organiser invites members via link or in-app invite. 2. Each member submits interests, budget and constraints. 3. System generates one reconciled itinerary, flagging conflicts and how each was resolved. 4. Members view the shared plan and propose changes visible to the whole group.
Alternative Flow	A member has not responded by a set point → organiser proceeds; the plan shows whose input was included.
Exception Flow	Preferences are irreconcilable within budget/time → system presents the closest compromise and explains the constraint.
Postconditions	A single shared itinerary exists that every invited member can see and was fairly derived from group input.

UC-05 — Trip Memory Box Management

Actor	Traveller
Goal	Preserve and later revisit trip photos and notes, organised by trip.
Preconditions	A trip exists (active or completed).
Trigger	User uploads a photo/note, or opens a past trip's memory box.
Main Flow	1. User uploads photo(s)/notes to a trip's memory box. 2. System stores items in encrypted, trip-linked storage. 3. User revisits the memory box at any time to view items. 4. Ahead of the retention limit, the system reminds the user with a one-tap download/extend option.
Alternative Flow	User downloads the full memory box in one action rather than item-by-item.
Exception Flow	Upload fails → system retries or clearly reports the failure; nothing is silently lost.
Postconditions	Trip memories remain organised, accessible, and protected from silent, surprise deletion.

UC-06 — Local Phrase / Translation Assistance

Actor	Traveller
Goal	Communicate basic needs in a region where the local language is unfamiliar.
Preconditions	Destination/region is known.

Trigger	User opens the phrase assistant, or the system proactively surfaces it on entering a new linguistic region.
Main Flow	1. System determines the relevant local language(s) for the destination. 2. User browses/searches common phrase categories. 3. Phrase is shown in local script with a phonetic transcription. 4. (Phase 2) User speaks/types for live two-way translation.
Alternative Flow	Region has multiple relevant local languages → user selects which to display.
Exception Flow	No phrasebook content yet for the region → system states this clearly rather than showing a broken screen.
Postconditions	User can handle basic communication needs (directions, prices, essentials) with reasonable confidence.

19. AI Requirements

AI is the core of this product, not a bolt-on feature. This section defines how the AI layer should treat each input type, then explains which AI techniques are needed and why — deliberately avoiding buzzwords that don't map to a real product need.

19.1 How AI Should Use Each Input

Input	How AI Uses It
User preferences	Primary signal for itinerary content and tone; explicit statements always outrank inferred ones (Section 16).
Destination information	Grounds itinerary generation and narration in real, current places rather than generic suggestions.
Budget	Constrains itinerary generation; drives budget-fit flags and (Phase 2) smart swap suggestions.
Time	Drives pacing, sequencing, and how many stops are realistic per day.
Location (real-time)	Powers on-site recommendations, arrival detection, and re-centring when off-route.
Weather	Triggers re-adaptation for outdoor activities (FR-014).
Attractions / POIs	Base content for itinerary items and heritage narration, sourced from curated + API data.
Restaurants	Filtered recommendations aligned to preferences, budget, and proximity.
Transportation	Informs realistic travel-time estimates between stops.
Historical / cultural information	Grounds heritage narration (RAG) — never generated purely from unguided model memory for factual claims.
User feedback	Highest-quality personalisation signal; explicit corrections persist across sessions (Section 20).
Previous trips	Seeds recommendations for new trips and reduces cold-start reliance on onboarding alone.

19.2 AI Technologies and Why Each Is Needed

- **LLM (Large Language Model):** powers conversational planning, itinerary modification, narration generation, and Q&A — needed anywhere the product must understand or produce natural language.
- **RAG (Retrieval-Augmented Generation):** grounds heritage narration and factual answers in verified, curated reference content instead of the model's unguided memory, directly reducing the risk of confident-sounding but wrong historical claims.
- **Recommendation system (hybrid):** rule-based filters (budget, hours, distance) combined with embedding similarity and review/behavioural signals — needed so POI and restaurant suggestions are both relevant and valid.
- **Embeddings + vector database:** needed to power semantic search over heritage/POI content and to match free-text interests to relevant places, rather than relying on exact keyword matches.
- **Personalisation / context management:** needed to carry a user's profile and current trip context across conversational turns and across sessions, so the assistant doesn't “forget” earlier in the same conversation.
- **Prompt engineering & guardrails:** needed to keep narration factual, keep the assistant on-topic, prevent unsafe or irrelevant outputs, and maintain a consistent companion tone.
- **AI evaluation:** needed to measure narration accuracy and recommendation relevance and to catch regressions before release, using golden test sets and human review sampling — particularly for historical claims, where an error carries real reputational and trust cost (Section 34).

20. Personalization Engine

Personalisation combines explicit signals (what the user directly tells the app: onboarding answers, stated preferences, budget, pace) with implicit signals (what the user's behaviour shows: accepted or rejected suggestions, time spent on a recommendation, review sentiment, prior trip history). Recommendations should keep shifting as more signal accumulates — personalisation is treated as an ongoing relationship, not a one-time onboarding form (Section 5.3).

Cold-start handling: for a brand-new user, the engine leans more heavily on explicit onboarding answers plus curated/popular defaults. As a user completes more trips and gives more feedback, the engine leans progressively more on learned behaviour. This directly addresses the tension identified in Section 43 between wanting the app to “know the A-Z of the person” and not wanting a long, abandon-prone onboarding form.

Rule: an explicit user correction always takes precedence over an inferred preference, immediately and for future recommendations — not just for the current session (see Section 16, Business Rules).

21. Dynamic / Real-Time Travel Intelligence

The itinerary is treated as a living plan, not a static document. The system monitors for conditions that would make the existing plan suboptimal or invalid, and proposes — rather than silently applies — an adjustment.

21.1 Reactive Triggers

- Weather changes affecting an outdoor activity.
- A planned attraction closes or changes hours unexpectedly.
- Traffic or delays that would make the next stop unrealistic on time.
- The user goes off-route from the planned location.
- A planned activity is missed or skipped.
- Actual spend trends over the planned budget.
- The user's own schedule changes (e.g. a later start time).
- A direct user request (“I'm tired, something more relaxed please”).

21.2 Design Principle

Every adjustment is presented to the user for confirmation, with a short plain-language reason, before it takes effect. Notifications are kept concise and actionable rather than frequent — the goal is an itinerary that adapts intelligently, not an app that interrupts constantly.

22. Tourism, Heritage & Culture

This is the product's most distinctive capability and deserves particular care, since the founder's own motivating example — a costly human guide at a site like the Taj Mahal — is a Tourism, Heritage & Culture problem specifically, not a generic travel-planning one.

- **Explain why, not just what:** narration should convey the significance of a place — why it matters historically, architecturally, or culturally — not just list facts a plaque already shows.
- **Layered storytelling:** a short factual layer for quick consumption, with a deeper narrative layer available on request for enthusiasts (ties to the Heritage Enthusiast persona, Section 9).
- **Cultural etiquette:** practical guidance — dress codes, customs, photography restrictions — surfaced at the right moment, not buried in a separate guide.
- **Beyond the obvious landmarks:** surfacing lesser-known local experiences alongside major sites, not only the standard “top 10” list (see Section 44, Hidden/Local Experiences).
- **Responsible tourism:** encouraging respectful behaviour at heritage sites and, where relevant, highlighting ways to support local communities and artisans.
- **Verified sourcing:** factual and historical content is sourced from vetted references (e.g. official tourism/heritage body materials, licensed content) rather than generated unguided by the model — this is what the RAG requirement in Section 19 exists to enforce.

23. Data Requirements

Data Type	What / Why	Storage & Retention
Profile data	Name, interests, travel style — core personalisation input.	Retained while account is active.
Real-time location	Powers on-site recommendations and arrival detection during an active trip.	Retained only for active-trip context; not kept long-term (Section 27).
Photo / memory data	User-uploaded trip photos and notes (memory box).	Default 12 months from upload, downloadable/extendable, reminder before expiry (Section 43).
Feedback / review data	Post-trip feedback (private) and public reviews.	Retained per standard product data policy; used to improve personalisation and public trust content.
Booking data	Records of link-outs and (later) in-app bookings.	Retained per standard recordkeeping norms — TBC pending legal/financial input.
Historical / cultural content	Curated, versioned reference content powering narration.	Versioned and refreshed on a defined content-review cadence (Section 36).
Interaction / usage data	Product analytics (funnels, feature usage).	Aggregated/anonymised where possible for analytics purposes.

As a matter of professional judgement beyond the founder's original notes: real-time location should not be retained indefinitely once a trip ends — only what is needed for that trip's context. This protects user trust and reduces the impact of any future data-security incident (see Section 27, Section 38).

24. API & Third-Party Integrations

Integration	Purpose	Phase	Failure Handling / Fallback
Google Maps Platform	Maps, Places, Directions, geocoding.	MVP	Cache last-known map data; degrade to list view if map tiles fail to load.
Google OAuth	Gmail-based sign-in.	MVP	Clear retry; no partial account created on failure.
Weather API	Forecast-aware planning and re-adaptation.	MVP	If unavailable, skip weather-based flags rather than blocking planning.
LLM / AI provider	Conversational planning, narration generation, Q&A.	MVP	Fallback to curated templates/cached answers; visible “limited mode” notice.
Translation API (text)	Basic phrasebook content.	MVP	Serve pre-downloaded phrase content offline if the API is unreachable.
Translation (live speech)	Two-way conversational translation.	Phase 2/3	N/A pre-launch; MVP relies on text phrasebook only.
Hotel booking (link-out)	Send users to partner sites to book.	MVP	If a partner link fails, offer an alternative partner or a manual search link.
Hotel/flight booking (API)	In-app automated booking.	Phase 2 / Future	N/A pre-launch.
Places / POI data provider	Attraction and restaurant data.	MVP	Fall back to curated local dataset if the live API is unavailable.
Push notification service (FCM)	Arrival, disruption, and reminder notifications.	MVP	Queue and retry; in-app notification centre as a backup channel.
Cloud object storage	Memory box photo/video storage.	MVP	Retry with backoff; user-facing upload-failed state, never silent loss.
Payments gateway	Required once in-app booking exists.	Future	N/A — no payment flows exist pre-booking phase.

25. UX/UI Requirements

The experience is conversational-first: chat is a primary way to interact with the product, not a hidden secondary feature. Two visually distinct modes reduce cognitive load — a Planning mode (research, comparison, editing) and an On-Trip mode (glanceable, location-aware, minimal typing) — rather than one interface trying to serve both moments equally well.

25.1 Key Screens

- Onboarding / interest quiz
- Home / dashboard (upcoming trip, quick actions, quick plan)
- Chat / AI companion interface
- Trip creation & itinerary view (map + timeline hybrid)
- On-trip / live companion mode (arrival notices, nearby suggestions, narration entry point)
- Heritage narration screen (layered content, photo Q&A entry point)
- Photo Q&A capture screen
- Budget view (planned vs actual)
- Memory box gallery (per trip)
- Collections / favourites
- Profile & preferences
- Notifications centre
- Reviews (view and submit)

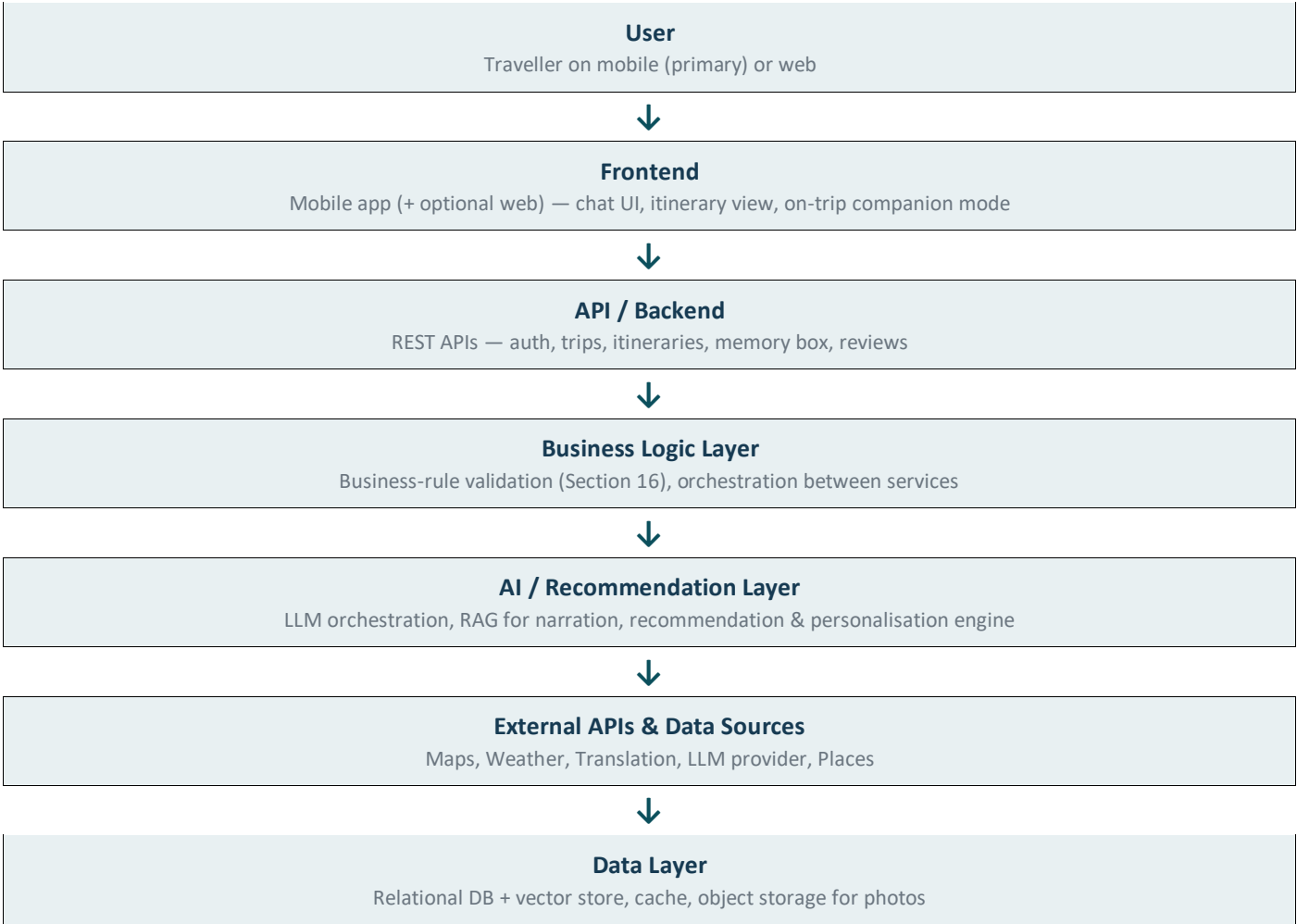
25.2 Experience Principles

- Map + timeline hybrid for itineraries, so spatial and time-based thinking are both supported.
- On-trip mode favours glanceable cards and minimal typing over dense text, since it is used while moving.
- Every AI-adjusted plan change is shown with a one-line reason, never applied silently (ties to Section 16, Section 21).
- Consent prompts (location, camera, notifications) explain the specific benefit unlocked, not just a generic permission request.

26. Technical Architecture

26.1 Layered Architecture

A conventional layered architecture is deliberately preferred over a more elaborate microservices setup — this is scoped to be realistic for a lean or student-scale team to actually build and operate (see Section 40, Constraints).



26.2 Authentication & Authorization

- Authentication: Google OAuth2 only for MVP (FR-002); no first-party password storage.
- Authorization: role-based access (traveller; internal/admin for content and moderation tooling).
- Session handling via short-lived access tokens plus refresh tokens, validated on every API call.

26.3 Key Data Flows

- **Itinerary generation flow:** client request → backend assembles user profile + trip parameters → AI/recommendation layer generates candidate plan → business-rule validation → response to client.
- **On-trip AI request flow:** location update → backend checks against active itinerary → triggers notification and/or fetches nearby recommendations → client displays.
- **Narration flow (RAG):** POI identified → relevant curated content retrieved from the vector store → LLM composes a grounded response → confidence-flagged output returned to client.

26.4 Recommended Technology Stack

“Recommended” reflects what is realistic for a lean team to build and actually ship; “Alternative” is a credible substitute if team composition or scale requirements change.

Layer	Recommended	Alternative
Mobile / Frontend	React Native (single codebase for Android/iOS)	Flutter; or React web app first, mobile later
Backend / API	Java + Spring Boot (REST APIs, Spring Security for OAuth2)	Node.js + Express, or Python + FastAPI
AI Orchestration	Backend service calling a hosted LLM API (no self-hosted models)	Dedicated Python microservice if deeper custom ML work emerges later
Database	PostgreSQL, with the pgvector extension for embeddings	Separate managed vector DB (e.g. Pinecone) at larger scale
Caching	Redis (session data, hot recommendation results)	In-memory cache within the API layer for the earliest MVP
Object Storage	S3-compatible storage for memory-box photos	Firebase Storage / Cloudinary
Notifications	Firebase Cloud Messaging (FCM)	OneSignal
Hosting	A managed cloud platform (containerised deployment)	PaaS (e.g. Render/Railway) for the earliest MVP to minimise DevOps overhead

This stack keeps the entire MVP realistically buildable by a small full-stack team without requiring a separate ML infrastructure specialism — the AI layer is orchestration (calling a hosted LLM and a vector-enabled Postgres store) rather than hosting or training models in-house.

27. Security & Privacy

- **Authentication:** Google OAuth2 only for MVP; no first-party password storage or management.
- **Authorization:** role-based access control; least-privilege access for internal/admin tooling.
- **Encryption:** TLS for all data in transit; encryption at rest for location history and memory-box photos.
- **API security:** rate limiting on all public endpoints, particularly AI endpoints (cost and abuse control).
- **Secret management:** API keys and credentials held in a secrets manager, never hardcoded or committed to source control.
- **Location privacy:** tracking is opt-in per trip, pausable/stoppable at any time, and retained only for active-trip context (Section 23).
- **User consent:** clear, specific consent flows for location and camera access, explaining the benefit unlocked (Section 25).
- **Data deletion:** users can request account and data deletion; memory-box items are never deleted silently (Section 43).
- **Payment security:** PCI-DSS-aligned processor required once payments are introduced (Future scope — not applicable pre-booking phase).
- **Audit logging:** sensitive actions (data export, account deletion, admin content edits) are logged for accountability.
- **AI data protection:** user data is not used to train third-party foundation models without explicit consent; narration content is sourced from vetted references, not arbitrary web scraping.

28. Edge Cases & Failure Handling

Scenario	Expected System Behaviour
Invalid/unknown destination	Prompt clarification; suggest nearest recognised matches rather than failing silently.
No recommendations available	Show a curated fallback list with an honest note; log the content gap for follow-up.
External API unavailable (Maps/Weather/LLM)	Degrade gracefully to cached/basic mode; show a status note; retry with backoff.
Attraction closed on planned day	Flag at generation time and again via dynamic monitoring; propose an alternative (FR-014).
User exceeds budget	Warn at generation time; allow an explicit override rather than silently blocking.
Conflicting preferences (group trip)	Present a reconciled option with the trade-off explained (Section 16).
Location unavailable / denied	Fall back to manual “I'm at ____” input; disable only the location-dependent features.
Internet unavailable	Serve cached itinerary and any pre-downloaded heritage/phrase content, read-only.
Booking failure (future)	Clear error message; no partial charge; offer retry or an alternative option.
Payment failure (future)	Standard failure handling; itinerary/trip data is never lost as a result.
AI hallucination / incorrect information	Confidence flagging; “verify locally” disclaimer on uncertain claims; user can report an issue.
User changes travel dates	Itinerary is re-validated (hours, weather, availability), not just copied to new dates.
User cancels a trip	Soft-delete/archive the trip; memory-box items already added are unaffected.

29. MVP Definition

The MVP is scoped to be realistic for a lean or student/hackathon-scale development team, per the instruction that not every idea belongs in the first release.

29.1 MVP (Phase 1)

- Google OAuth sign-in (FR-002)
- Onboarding interest/traveller-type capture (FR-003)
- AI itinerary generation — chat + structured input, including the user's own trip ideas (FR-001, FR-004, FR-005)
- Google Maps integration for viewing the itinerary (Module 9)
- Budget estimate at planning time (FR-016, tracking only, no splitting yet)
- Real-time location-based arrival notifications (FR-006)
- Heritage narration for a curated shortlist of 5–10 flagship points of interest (FR-007) — directly matching the founder's own suggestion to “try implementing for only a few places first”
- Basic text phrasebook, not live speech translation (FR-009)
- Memory box: photo/notes upload, view, download (FR-010)
- Trips list, favourites/collections (FR-011)
- Basic reviews — submit and view (FR-013)

29.2 Phase 2

- Group/collaborative trip planning (FR-012)
- Dynamic itinerary re-adaptation (FR-014)
- Safety/SOS trusted-contact sharing (FR-017)
- Weekend/local outing Quick Plan mode (FR-015)
- Budget tracking with group expense splitting (extends FR-016)
- Expanded heritage POI catalog beyond the initial shortlist
- Live speech translation, via a licensed translation API

29.3 Phase 3 / Advanced

- Automated hotel/flight booking via partner APIs (BR-013)
- Offline mode for maps and heritage content
- AI-generated trip recap / shareable memory summaries
- Predictive, proactive recommendations
- “What if?” trip simulation
- Richer, model-assisted group preference reconciliation

30. Feature Prioritization

MoSCoW prioritisation applied to the modules defined in Section 12.

Priority	Modules	Reasoning
Must Have	Auth, Onboarding, AI Planner, Itinerary Engine, Maps, Location Companion, Heritage Narration (curated), Memory Box, Trip Management	These are the modules the core value proposition cannot function without.
Should Have	Visual Q&A, Phrasebook, Attractions/Restaurants, Collections & Favourites, Reviews, Notifications, Analytics	Materially strengthen the MVP but the product is still coherent without them on day one.
Could Have	Budget Tracker (live), Weekend/Local Outing Planner, Collaborative Planning, Safety/SOS	High value but reasonably deferred to Phase 2 without weakening the initial value proposition.
Won't Have Yet	Automated hotel/flight booking, live speech translation, full-catalog heritage coverage	Require commercial API agreements, payment infrastructure, or content-scale investment not realistic pre-MVP (Section 43).

31. Product Roadmap

Phase 1 — MVP: Core Personalised Travel Planning

Establish the core loop: personalised planning, on-site companion guidance, curated heritage narration, and trip memories — proving the central thesis with a lean, realistic feature set (Section 29.1).

Phase 2 — Intelligent Travel Companion

Conversational modification matures, dynamic itinerary adaptation ships, group/collaborative planning launches, and the phrase assistant grows toward live translation.

Phase 3 — Travel Ecosystem

Booking integrations and payments connect the product to real transactions; partner network (hotels, experiences) expands; offline mode ships.

Phase 4 — Advanced AI Companion

Predictive and proactive capabilities mature: AI trip-recap generation, “what if?” simulation, and a substantially expanded heritage content catalog via a repeatable content pipeline.

32. Complete SDLC

An Agile/Scrum approach is used throughout (Section 33); these eleven phases describe the full lifecycle, not a single rigid waterfall pass — phases 6–11 repeat every sprint/release cycle.

Phase 1 — Requirement Gathering

Activities	Capture the founder's vision and raw ideas; interview stakeholders where available; review comparable products.
Deliverables	Raw requirements notes; this master document's early sections (2–11).
Responsible Roles	Product Owner, Business Analyst
Entry Criteria	Product idea exists (the founder's brainstorming document).
Exit Criteria	Business context, problem statement and product scope are documented and agreed.

Phase 2 — Requirement Analysis

Activities	Break down business requirements into functional and non-functional requirements; identify gaps, risks and assumptions.
Deliverables	Sections 13–16, 38–41 of this document.
Responsible Roles	Business Analyst, Product Manager
Entry Criteria	Business requirements are documented.
Exit Criteria	Functional/non-functional requirements are specific, testable and reviewed.

Phase 3 — Product Planning

Activities	Define MVP scope, prioritise features, sequence the roadmap.
Deliverables	Sections 29–31 (MVP Definition, Prioritisation, Roadmap).
Responsible Roles	Product Manager, Product Owner
Entry Criteria	Requirements are analysed and stable enough to scope.
Exit Criteria	MVP scope, priorities and roadmap are agreed and documented.

Phase 4 — UX/UI Design

Activities	Design core flows and screens; validate the conversational-first, dual-mode (planning vs on-trip) approach.
Deliverables	Wireframes/prototypes for key screens (Section 25); design system basics.
Responsible Roles	UI/UX Designer, Product Manager
Entry Criteria	MVP scope is defined.
Exit Criteria	Core flows are designed and validated (at minimum via internal review; usability testing where feasible).

Phase 5 — System Architecture

Activities	Finalise the layered architecture, technology stack, and integration list.
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Deliverables	Section 26 (Technical Architecture), Section 24 (Integrations).
Responsible Roles	Solution Architect, Backend/AI Engineers
Entry Criteria	Functional requirements and NFRs are stable.
Exit Criteria	Architecture and stack are agreed; API contracts for MVP modules are outlined.

Phase 6 — Development

Activities	Build MVP modules in prioritised order (Section 30); implement business rules (Section 16); integrate external APIs.
Deliverables	Working software increments, per sprint.
Responsible Roles	Frontend/Backend/AI Engineers
Entry Criteria	Architecture is agreed; designs for the sprint's scope are ready.
Exit Criteria	Features meet acceptance criteria (Section 14, Section 17) and pass code review.

Phase 7 — Testing

Activities	Execute the testing strategy (Section 34) across functional, non-functional, and AI-response dimensions.
Deliverables	Test reports, defect log, AI evaluation results.
Responsible Roles	QA Engineer, AI Engineer
Entry Criteria	A testable build is available.
Exit Criteria	Critical/high defects resolved; AI narration accuracy meets the agreed bar for launch POIs.

Phase 8 — Deployment

Activities	Release to staging, run final checks, deploy to production following the CI/CD process (Section 35).
Deliverables	Deployed production release; release notes.
Responsible Roles	DevOps Engineer, Backend Engineers
Entry Criteria	Build has passed QA sign-off.
Exit Criteria	Release is live in production and passes smoke tests.

Phase 9 — Monitoring

Activities	Track application, API, AI-quality and security signals (Section 36).
Deliverables	Dashboards, alerts, incident log.
Responsible Roles	DevOps Engineer, Product Manager
Entry Criteria	Release is live.
Exit Criteria	Monitoring is actively catching issues before most users report them.

Phase 10 — Maintenance

Activities	Fix defects, update dependencies, refresh heritage content, tune AI prompts/guardrails based on real usage.
Deliverables	Patch releases; content-refresh log.
Responsible Roles	All engineering roles
Entry Criteria	Product is live with real users.
Exit Criteria	Ongoing — recurring cadence, not a one-time exit.

Phase 11 — Continuous Improvement

Activities	Review KPIs (Section 37) against goals; feed learnings back into the roadmap (Section 31) and personalisation engine (Section 20).
Deliverables	Periodic product-review reports; updated roadmap.
Responsible Roles	Product Manager, Product Owner
Entry Criteria	Sufficient usage data has accumulated.
Exit Criteria	Ongoing — feeds back into Phase 1–3 for the next cycle.

33. Development Methodology

Agile/Scrum is recommended over a waterfall approach, given the product's evolving AI-driven uncertainty and the realities of a small or student-scale team — short feedback loops matter more here than upfront certainty.

- **Sprint structure:** 2-week sprints, which is enough time to ship a coherent slice of a module without losing momentum on a small team.
- **Backlog:** seeded directly from this document's user stories (Section 17), prioritised per Section 30.
- **Sprint planning:** select the next slice of Must/Should items from the MVP definition (Section 29), sized to the team's real velocity.
- **Development:** trunk-based or short-lived feature branches, to avoid long-lived branches diverging in a small team.
- **Code review:** every change reviewed via pull request before merge; no direct pushes to the main branch.
- **Testing:** automated tests run in CI before merge (Section 34); manual exploratory testing for AI-response quality.
- **Sprint review:** demo completed work against the sprint's acceptance criteria.
- **Retrospective:** a short “what worked / what didn't” each sprint — particularly important early on, while the AI layer's real-world behaviour is still being learned.
- **Release process:** staged promotion — dev → staging → production — with no direct-to-production deploys (Section 35).

34. Testing Strategy

Testing spans conventional software quality dimensions plus a dimension unique to this product: evaluating whether AI-generated content (itineraries, narration, answers) is actually correct, not just whether the code runs without error.

Test Type	Focus	Suggested Tooling
Unit testing	Business-rule logic (Section 16), request/response mapping, utility functions.	JUnit 5 + Mockito (Java/Spring stack)
Integration testing	Service-to-service and service-to-database interactions.	Spring Boot Test + Testcontainers
API testing	Contract correctness, status codes, error handling for every FR endpoint.	RestAssured; Postman/Newman for collections
UI testing	Critical mobile flows — onboarding, itinerary creation, on-trip mode.	Espresso/XCUITest, or Appium for cross-platform
System / E2E testing	Full user journeys end-to-end (Section 10), across real integrations in staging.	Playwright (web) / Appium (mobile)
Regression testing	Re-run on every release to catch breakage in previously working features.	Automated suite in CI; tagged smoke subset for fast checks
Performance testing	Itinerary-generation latency and API throughput under load (Section 15 targets).	JMeter or k6
Security testing	Auth flows, input validation, dependency vulnerabilities.	OWASP ZAP baseline scan; automated dependency scanning in CI
Usability testing	Onboarding completion and core-flow comprehension with real users.	Moderated sessions with a small user panel
User Acceptance Testing	Stakeholder sign-off against the acceptance criteria in Sections 14 and 17.	Structured UAT checklist mapped to Section 42
AI response evaluation	Narration accuracy, recommendation relevance, hallucination rate.	Golden test sets + human review sampling; adversarial prompt tests for guardrails

The AI response evaluation row is the least “off-the-shelf” part of this suite and deserves the same rigor as the automated functional suite: a small, versioned “golden set” of POIs with known-correct narration facts, re-run on every change to prompts or retrieval content, catches silent regressions that ordinary functional tests will not.

35. Deployment & DevOps

- **Environments:** Development → Testing → Staging → Production, with no direct-to-production changes.
- **CI/CD:** GitHub Actions (realistic free-tier fit for a student/lean team) running build, lint, and the automated test suite on every pull request.
- **Environment variables & secrets:** never hardcoded or committed; managed via the platform's secret store or a dedicated secrets manager.
- **Database migrations:** versioned migrations (e.g. Flyway, fitting the recommended Java/Spring stack) applied automatically as part of deployment, never manually against production.
- **Monitoring & logging:** see Section 36.
- **Rollback:** versioned releases with a simple redeploy-previous-version rollback path — blue-green deployment is a reasonable later upgrade once traffic justifies it.
- **Backups:** automated daily database backups at minimum; retention period TBC (Section 15, Disaster Recovery).

36. Monitoring & Maintenance

- **Application monitoring:** uptime and error-rate tracking for core services.
- **API monitoring:** latency and error rate per external integration (Section 24), so a degraded provider is caught before users report it.
- **Error monitoring:** centralised error capture (e.g. Sentry or equivalent) across frontend and backend.
- **AI quality monitoring:** hallucination-flag rate, “report an issue” volume on narration, and narration-confidence trend over time.
- **Performance monitoring:** response-time percentiles against the targets in Section 15.
- **User analytics:** funnel tracking from onboarding → itinerary generated → trip started → trip completed.
- **Security monitoring:** authentication failure patterns and anomalous access.
- **Incident management:** lightweight severity levels and a clear triage owner — sized for a small team, not an enterprise on-call rotation.
- **Maintenance process:** regular dependency updates and a defined cadence for reviewing heritage-content freshness and accuracy.

37. KPIs & Success Metrics

Metric	What It Tells Us
Itinerary generation success rate	How often the AI planner produces a usable itinerary without failing or falling back.
Recommendation acceptance rate	Share of suggestions accepted vs modified/rejected — a direct personalisation-quality signal.
User satisfaction (post-trip rating)	Overall experience quality, captured via FR-018.
Trip completion rate	Share of planned trips actually taken — a proxy for plan realism and trust.
Itinerary modification rate	How often users need to correct the AI's first attempt — lower is better over time.
Engagement across trip lifecycle	Sessions during planning vs. on-trip vs. post-trip — tests the “companion, not one-time tool” thesis.
Retention / repeat trip creation	Whether users come back for a second trip — the clearest test of the companion relationship.
Narration engagement rate	Share of visits to a supported POI where narration was actually opened.
Memory box adoption rate	Photos/notes uploaded per completed trip — tests whether the memory box earns a place in the user's habits.
Booking conversion	Once booking integration exists: link-outs or in-app bookings completed.
AI response quality (accuracy sampling)	Human-reviewed sample accuracy score for narration and Q&A answers.

38. Risks

Risk	Prob.	Impact	Mitigation	Owner
AI hallucination in heritage narration	Medium	High	RAG grounding, guardrails, confidence flagging, human content review for launch POIs (Section 19, 22).	AI Eng
Third-party API outage (Maps/LLM/Weather)	Medium	Medium	Graceful degradation, caching, provider fallback where feasible (Section 24).	Backend Eng
Data privacy / location misuse concerns	Low	High	Privacy-by-design, minimal retention, explicit consent (Section 27).	Product Owner
LLM/API cost overrun at scale	Medium	Medium	Usage caps, caching, prompt/response size optimisation, cost monitoring.	Eng Lead
Content accuracy / liability for historical claims	Low	High	Vetted sources only, clear disclaimers on uncertain claims (Section 22).	Content/AI Eng
Scope creep dilutes core product (e.g. dating features)	Medium	Medium	Strict MVP scope discipline; phased roadmap (Section 43).	Product Owner
Low initial content coverage limits perceived value	Medium	Medium	Realistic, expanding curated-POI roadmap rather than an overpromised launch catalog.	Product/Content
Small-team bandwidth constraints	High	Medium	Realistic MVP scoping; avoid over-engineering (Section 29, 40).	Product Owner
Booking-partner dependency for monetisation	Low	Medium	Diversify partners; don't block core value on booking revenue.	Product Owner
Security incident (account/photo data breach)	Low	High	Standard security controls, incident response plan (Section 27).	DevOps/Eng Lead
Competitive response from larger travel platforms	Medium	Medium	Focus on the defensible niche — on-site companion + domestic-language heritage storytelling — rather than competing on booking price.	Product Owner

39. Assumptions

Stated explicitly as assumptions, not facts, per the instruction not to present them as settled:

- The primary initial market is domestic Indian travel.
- Users have smartphones with GPS, a camera, and reasonable data connectivity during trips (with offline access addressing partial gaps in a later phase).
- Dependence on the Google ecosystem (Maps, Gmail OAuth) is acceptable for the MVP.
- A small or student-scale team is building the MVP, which shapes the architecture and scope recommendations throughout this document.
- Monetisation is not required to validate the MVP; the initial goal is proving the product experience.
- Heritage/historical content for the initial curated POI shortlist can be sourced and verified within reasonable effort.
- Users are willing to grant location and camera permissions in exchange for the core value delivered.
- All numeric targets in this document (performance, uptime, retention) are Proposed Targets, not committed SLAs, until validated against real usage.

40. Constraints

- **Budget:** limited / self-funded or hackathon-scale budget assumed — TBC.
- **Team size:** small/student team assumed, which directly shaped the architecture recommendation in Section 26.
- **Time:** hackathon/academic timeline assumed — exact duration TBC.
- **API limitations:** free-tier quotas on Maps, LLM, and translation APIs will likely cap early usage volume.
- **Technology limitations:** mobile OS background-location permission models vary by platform and constrain always-on tracking.
- **Data limitations:** authoritative heritage content is not equally available for every monument, which is exactly why MVP starts with a curated shortlist.
- **AI limitations:** LLMs can hallucinate, carry a training-data cutoff, and cannot guarantee real-time factual accuracy without grounding (Section 19).

41. Dependencies

41.1 Internal Dependencies

- A basic design system/UI kit must exist before frontend development can move quickly.
- Backend APIs must be ready (or stubbed) before frontend integration work can proceed in parallel.
- Curated heritage content must be prepared and verified before the narration feature can ship for a given POI.

41.2 External Dependencies

- Google Maps Platform and Google OAuth.
- A weather data provider.
- An LLM/AI provider.
- A translation API/service.
- A push notification service (FCM).
- Cloud hosting and object storage providers.
- Future: hotel/flight booking partners and a payments gateway.

42. Requirement Traceability Matrix

Each business requirement traces through to its functional requirement(s), a representative user story, acceptance criteria, and a test scenario. Where a requirement is cross-cutting or architectural rather than user-facing, the User Story column notes this rather than forcing an artificial story reference.

BR	FR(s)	User Story	Acceptance Criteria (brief)	Test Scenario
BR-001	FR-001, FR-004, FR-005	US-001, US-002, US-003	Itinerary matches stated interests/budget/time/location and reflects the user's own ideas.	Itinerary-generation functional & API test.
BR-002	FR-002	(FR-level AC — no dedicated story)	User signs in via Google; no partial account created if consent is denied.	OAuth sign-in functional test.
BR-003	FR-006	US-005, US-006	Arrival notification and nearby suggestions delivered within target latency.	Geofence/notification functional test.
BR-004	FR-007	US-008	Grounded narration loads within target time for a supported POI.	Narration content regression test (golden set).
BR-005	FR-008	US-009	Relevant, grounded answer returned; low-confidence answers visibly flagged.	Visual Q&A functional test.
BR-006	FR-009	US-010	Phrases shown with transliteration; available offline once downloaded.	Phrasebook functional + offline test.
BR-007	FR-010	US-011, US-012	Uploaded item appears immediately; expiry reminder sent before deletion.	Upload / download / retention test.
BR-008	FR-011	US-019	Favourited item appears under Favourites immediately.	Favourites CRUD test.
BR-009	FR-012	US-013, US-014	Conflicting group preferences are flagged with the resolution shown.	Multi-user collaboration test.
BR-010	FR-013	US-017, US-018	Submitted review appears after passing moderation.	Review CRUD / moderation test.
BR-011	FR-015	US-015	Short plan (1–3 stops) returned for given time/budget/occasion.	Quick-Plan functional test.
BR-012	FR-005	US-002	Places named in pasted notes appear in the generated plan.	Idea-extraction functional test.
BR-013	(Architecture — Sec. 24, 26)	N/A — architectural requirement	Booking module interface is documented/stubbed for future use.	Architecture / interface-contract review.
BR-014	FR-006, FR-010	N/A — cross-cutting privacy requirement	Explicit consent required before location or camera access is used.	Privacy/consent-flow test + security review.
BR-015	FR-017	US-016	Trusted contact sees live location; SOS notifies within target latency.	SOS / location-sharing functional test.
BR-016	FR-018	US-018	Next itinerary reflects at least one piece of prior feedback.	Personalisation regression test.

BR	FR(s)	User Story	Acceptance Criteria (brief)	Test Scenario
BR-017	FR-014	US-007	Adjustment proposed before the affected activity's scheduled time; dismissal preserves the original plan.	Disruption-simulation functional test.
BR-018	FR-016	(FR-level AC — no dedicated story)	Actual-spend total updates immediately; over-budget notice triggers correctly.	Budget-tracking functional test.
BR-019	FR-003	US-004	First itinerary reflects at least one interest selected during onboarding.	Onboarding functional test.

43. Product Review & Recommendations

A direct, critical review of the founder's original ideas, as requested — including where an idea is strong as-is, where it needs adjustment, and why.

43.1 Full Entrance-to-Exit Narration for Every Monument

Original Idea	On entering a building like the Taj Mahal, full history should be available from entrance to exit — acknowledged by the founder as not fully practical, with a suggestion to start with only a few places.
Assessment	Directionally excellent — arguably the product's single most differentiated idea — but genuinely not feasible as “every place” for MVP or, realistically, ever at unlimited scale. Content creation, verification and licensing effort scale directly with the number of POIs covered.
Recommended Change	Launch with a curated shortlist of 5–10 flagship POIs (the Taj Mahal included) with genuinely deep, verified narration, then scale via a repeatable content pipeline rather than promising national coverage upfront.
Reason	Protects narration quality and trust — a confidently wrong historical claim does more damage than an honestly missing one — while still proving the core concept end-to-end.

43.2 Dating Planner

Original Idea	The app should also work as a dating planner based on location, time and budget, alongside weekend and friends-outing planning.
Assessment	An interesting extension of the personalisation engine, but a genuine brand and trust risk if bundled into the same product as a travel/heritage companion — dating features carry different audience expectations and different safety/moderation requirements (e.g. meeting strangers) than travelling with an AI guide.
Recommended Change	Keep the underlying planning engine reusable for occasion-based local outings broadly (dates, friend hangouts, family outings), but do not build dating-specific matching or discovery features into this product. If pursued, treat it as a separate spin-off sharing the recommendation engine.
Reason	Avoids scope creep and avoids importing a different product's trust-and-safety requirements into a heritage/travel-companion brand (see Section 38, Scope-creep risk).

43.3 Memory Box Retention — “Up to One Year”

Original Idea	Trip photos are stored in a per-trip memory box, kept for up to one year, downloadable or viewable any time.
Assessment	A reasonable default, but as originally stated it leaves an important gap: what actually happens at the 12-month mark — silent deletion, archiving, or a prompt to renew?
Recommended Change	Default 12-month active retention, with a reminder notification before expiry and a one-tap “download all” or “extend” action — never silent deletion.
Reason	The founder's own framing promised users a secure space they “need not worry about” — that promise requires a policy that never surprises the user with unannounced data loss.

43.4 Live Speech Translation

Original Idea	The app should hear the local language and convert it to a language the user knows, similar to existing live-translation apps.
Assessment	High user value, but technically demanding at launch — accurate real-time speech recognition and translation across many Indian regional languages, dialects and accents is a substantial engineering and cost commitment.

Recommended Change	MVP ships text-based common-phrase assistance (a curated, offline-capable phrasebook); live conversational speech translation becomes a Phase 2/3 capability, most realistically via a licensed translation API rather than building recognition/translation in-house.
Reason	Keeps the MVP realistic for a lean team (Section 40) while fully preserving the long-term vision on the roadmap (Section 31).

43.5 The App Should Know the “A–Z” of the Person

Original Idea	The application should know everything about the person using it and act based on their interests and preferences — essentially a full companion.
Assessment	The right long-term direction, but demanding this all upfront creates a real cold-start and onboarding-abandonment risk — nobody completes a long questionnaire before their first itinerary.
Recommended Change	Progressive profiling: a short onboarding quiz captures the essentials, and the profile deepens through actual usage — accepted/rejected suggestions, reviews, explicit corrections — rather than one long upfront form.
Reason	Balances personalisation depth against onboarding completion rate (Section 15, Usability; Section 20, Personalisation Engine).

43.6 Weekend Planning, Friends Outings — “Not Only a Travel Planner”

Original Idea	Beyond multi-day trips, the app should personally help plan weekend outings and friend hangouts based on location, time and budget.
Assessment	A good instinct — it raises engagement frequency between big trips — but risks diluting the core “tourist guide” identity if built out fully alongside everything else at once.
Recommended Change	Ship the core travel-companion experience first. Introduce a lightweight “Local Plans” / Quick Plan mode (FR-015) as a Phase 2 feature, explicitly scoped smaller than full trip planning — no heritage narration, no multi-day logistics.
Reason	Keeps MVP focused (Section 29) while preserving the broader vision explicitly on the roadmap rather than dropping it.

43.7 Automatic Hotel & Flight Booking via APIs

Original Idea	In future, automatic hotel and flight booking could be added by integrating booking-site APIs.
Assessment	Correctly self-deferred by the founder already — this is a heavy commercial and legal undertaking (payment handling, inventory contracts, cancellation policy) that would be premature pre-MVP.
Recommended Change	Design data models and module boundaries now so booking can be added later without a rebuild (a stubbed Booking module interface, BR-013), but do not attempt real booking integration before the MVP is validated.
Reason	Matches BR-013 and avoids the cost/scope-overrun risk flagged in Section 38.

43.8 The iPhone Analogy — Combining Existing Capabilities

Original Idea	Like a smartphone combined many separately invented technologies into one product, this app can combine existing capabilities (translation, maps, AI chat) into one integrated product.
Assessment	Correct and strategically important framing. The product's differentiation is integration and companion experience, not inventing new base technology — maps, translation and LLMs already exist as commodities.

Recommended Change	No change needed to the idea itself. It is adopted directly as the product's strategic thesis (Section 5) so the team focuses engineering effort on orchestration, companion UX and heritage content — not on reinventing commodity infrastructure like a custom translation engine.
Reason	Keeps build effort aimed at the actual differentiator instead of low-value “not invented here” engineering.

44. Innovation Opportunities

Several candidate innovations are already folded into the product as scoped above; they are marked “Already incorporated” below rather than re-argued. The remainder are evaluated on their own merits.

Innovation	User Value	Business Value	Complexity	MVP Suitability
Dynamic itinerary regeneration	High — plans stay trustworthy as conditions change.	Higher retention through disruption.	Medium	Already incorporated — FR-014, Phase 2
Context-aware AI companion	High — the core product thesis.	Primary differentiator.	Medium–High	Already incorporated — core MVP vision
Cultural storytelling	High — directly answers the founder's guide-cost motivation.	Strong brand differentiation.	Medium	Already incorporated — Section 22, MVP
Group preference reconciliation	High for group travellers.	Expands addressable use cases.	Medium–High	Already incorporated — FR-012, Phase 2
Weather-aware planning	Medium–High — avoids ruined outdoor plans.	Reduces poor-experience complaints.	Low–Medium	Already incorporated — FR-014, MVP/Phase 2
Smart budget optimisation	Medium — suggests cost-saving swaps within budget.	Differentiates from static budget estimators.	Medium	Phase 2
Crowd-aware recommendations	Medium — avoids peak-crowd times/places.	Nice-to-have differentiator.	High (needs real-time crowd data)	Phase 3 — data availability risk
Travel fatigue-aware scheduling	Medium–High, especially for family/elderly travellers.	Strengthens the Family persona's fit.	Medium	Phase 2/3
Hidden / local experiences	High — differentiates from generic “top 10” lists.	Enables local-partner relationships.	Medium	Phase 2
Responsible tourism prompts	Medium — etiquette and visitor-load awareness.	Brand and goodwill value.	Low–Medium	Phase 2
Offline AI travel assistance	High at low-connectivity heritage sites.	Protects the core differentiator's reliability.	Medium–High	Phase 2/3 — see Section 12 Recommended Enhancement
AI travel memory (auto trip recap)	Medium–High — shareable, emotionally resonant.	Potential viral/sharing loop.	Medium	Phase 2/3
“What if?” trip simulation	Medium — fun, exploratory planning aid.	Engagement novelty; lower urgency.	Medium–High	Phase 3 — lower priority
Predictive recommendations	High once mature, low initially.	Long-term retention driver.	High (needs substantial usage data first)	Phase 3/4

Closing Note

This document intentionally leaves several fields marked TBC or Proposed Target rather than inventing precision that does not yet exist — team size, exact budget, committed SLAs, and hard market-size figures should be filled in as real constraints and real usage data become available, not assumed at the outset. Everything else here is intended to be immediately actionable: a Product Manager can run Section 30 as a backlog, a Developer can build directly from Sections 14 and 26, a QA Engineer can derive test cases directly from Sections 14, 17 and 34, and a Designer can start from Section 25 and the personas in Section 9.